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MULTIMODAL DEEP LEARNING FRAMEWORK FOR BRAIN TUMOR PREDICTION

A Graduation Project (2) Prepared to Obtain the Degree of B.Sc. in the Department of Artificial Intelligence Engineering and Data Science

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SUPERVISOR CERTIFICATION

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الملخص

تعتبر أورام الدماغ مثل الأورام الدبقية (Gliomas) من أخطر الأمراض المهددة للحياة، ويعد التشخيص المبكر والدقيق وتحديد درجة الورم أمراً حاسماً لإنقاذ حياة المرضى وتوجيه الخطط العلاجية. ومع ذلك، تواجه الأنظمة المؤتمتة الحالية تحديات تتعلق بضعف موثوقية النماذج عند تطبيقها سريرياً، بالإضافة إلى محدودية الاعتماد على نمط تصوير طبي واحد، مما قد يغفل تفاصيل حيوية متكاملة لخصائص الأنسجة المصابة.

يتضمن هذا البحث مساهمة ثنائية؛ تبدأ أولاً بإجراء دراسة مرجعية منهجية حديثة (State-of-the-art SLR) لتهيئة أساس نظري متين وتحديد الفجوات البحثية بدقة. بناءً عليها، يقدم البحث إطار عمل متطور يعتمد على التعلم العميق متعدد الأنماط (Multimodal Deep Learning) لدمج عدة أنماط من صور الرنين المغناطيسي ثلاثية الأبعاد مثل .(T1, T1CE, T2, FLAIR) تكمن القيمة المضافة للنظام في تبني نموذج تجميعي ثلاثي الأبعاد (3D Ensemble) يخضع لآليات تحكم صارمة لمنع تسرب البيانات(Leakage-Controlled) ، مما يضمن تقييماً موثوقاً وعالي الدقة لدرجات الأورام ويبرر كفاءة دمج هذه التقنيات لخدمة القرار الطبي الذكي.

أثبتت النتائج التجريبية على مجموعة بيانات BraTS 2018 كفاءة إطار العمل التجميعي المقترح، حيث حقق النموذج متوسط دقة تصنيف (Accuracy) بلغ 88.07%، ومعيار F1-score بنسبة 91.95%، ومساحة تحت المنحنى (AUC) بمقدار 0.9114. ووصفياً، تميز النظام بتحقيق توازن مثالي بين الإيجابيات الخاطئة والسلبيات الخاطئة (بمعدل 3.4 لكل طية)، كما أدت عملية معايرة الاحتمالات (Platt Calibration) إلى تحسين الموثوقية السريرية بشكل ملحوظ عبر خفض خطأ المعايرة المتوقع (ECE) بنسبة 26.9% وخفض درجة Brier بنسبة 16.8%. يُستنتج من ذلك أن توفير نقاط تشغيل سريرية مرنة وقابلة للتحكم (مثل عتبة الحساسية العالية التي تحقق معدل استدعاء 95.24% مع 3 سلبيات خاطئة فقط) يجعل من هذا الإطار نظاماً موثوقاً، متوازناً، وجاهزاً للنشر لدعم اتخاذ القرار الطبي في تشخيص الأورام.

كلمات مفتاحية: التعلم العميق، الأنظمة متعددة الأنماط، التنبؤ بأورام الدماغ، الأورام الدبقية، الموثوقية السريرية، مراجعة الأدبيات المنهجية(SLR) .

Abstract

Brain tumors, particularly gliomas, represent highly aggressive and life-threatening malignancies, where early detection and precise histological grading are critical for effective treatment planning and patient survival. Traditional computer-aided diagnosis systems face key challenges, primarily their heavy reliance on single-modality medical imaging which often misses complementary tissue characteristics, alongside substantial limitations regarding model reliability, generalizability, and data leakage when deployed in clinical environments.

To address these limitations, this thesis first conducts a state-of-the-art Systematic Literature Review (SLR) to thoroughly map the recent architectural landscapes and isolate existing research gaps. Consequently, we propose a robust Multimodal Deep Learning framework that integrates multiple 3D MRI sequences (including T1, T1CE, T2, and FLAIR). The core novelty of this framework lies in deploying a leakage-controlled 3D ensemble paradigm, which ensures absolute data separation, mitigates optimistic performance bias, and drastically improves prediction reliability—thereby justifying its application as a dependable second-opinion tool in neuro-oncology.

Experimental evaluation on the BraTS 2018 dataset demonstrated the high efficacy of the proposed ensemble framework, achieving an average Accuracy of 88.07%, an F1-score of 91.95%, and an Area Under the Curve (AUC) of 0.9114. Descriptively, the system showcased a perfectly balanced error profile with an equal False Negative to False Positive ratio (3.4 per fold). Furthermore, post-hoc probability calibration significantly improved clinical reliability by reducing the Expected Calibration Error (ECE) by 26.9% and the Brier score by 16.8%. In conclusion, by providing controllable clinical operating points (such as a high-sensitivity threshold yielding a 95.24% recall with only 3 false negatives), the framework establishes itself as a robust, calibration-aware, and highly deployable tool for medical decision support in neuro-oncology.

Keywords: Deep Learning, Multimodal Fusion, Brain Tumor Prediction, Glioma Grading, Clinical Reliability, Systematic Literature Review (SLR).

List of Publications

The following research papers have been submitted for publication based on the work and methodologies presented in this thesis:

Papers Under Review

- [1] M. A. Al-Saleh, R. Sanjakdar, and M. Aboukassem, "Reliable Glioma Grading via Leakage-Controlled Multimodal 3D Ensemble Learning," Submitted to *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI) Mentorship Program*, 2026. [Status: Under Review]
- [2] M. A. Al-Saleh, R. Sanjakdar, M. Aboukassem, and A. Abdulaziz, "From Accuracy to Clinical Reliability: A Systematic Review of Deep Learning in Glioma Grading," Submitted to *IEEE*, 2026. [Status: Under Review]

(Note: The title pages of these manuscripts are attached in Appendix A).

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Abbreviation	Abbreviation	المصطلح
AI	Artificial Intelligence	الذكاء الاصطناعي
DL	Deep Learning	التعلم العميق
MRI	Magnetic Resonance Imaging	التصوير بالرنين المغناطيسي
HGG	High-Grade Gliomas	الأورام الدبقية عالية الدرجة
LGG	Low-Grade Gliomas	الأورام الدبقية منخفضة الدرجة
D CNN3	D Convolutional Neural 3 Networks	الشبكات العصبية الالتفافية ثلاثية الأبعاد
ViT	Vision Transformers	محولات الرؤية
-	Self-Attention	الانتباه الذاتي
Swin	Shifted-Window Self-Attention	الانتباه الذاتي بالنافذة المزاحة
MIL	Multiple Instance Learning	التعلم متعدد الأمثلة
OOF	Out-of-Fold	تنبؤات خارج الطية
-	Platt Scaling	معايرة بلات
-	Shannon Entropy	إنتروبيا شانون
SLR	Systematic Literature Review	المراجعة المنهجية للأدبيات
ECE	Expected Calibration Error	خطأ المعايرة المتوقع
-	Brier Score	مقياس / درجة براير
FN	False Negatives	السلبيات الخاطئة
FP	False Positives	الإيجابيات الخاطئة
-	Accuracy	الدقة
-	Precision	الضبط
-	Recall / Sensitivity	الاستدعاء / الحساسية
-	Specificity	النوعية
-	F1-Score	درجة إف 1
AUC-ROC	Area Under the ROC Curve	المساحة تحت منحنى خصائص تشغيل المستقبِل
ROI	Region of Interest	منطقة الاهتمام
CV	Cross-Validation	التحقق المتقاطع
SOTA	State-of-the-Art	أحدث ما توصلت إليه التقنية (حالة الفن)
LLMs	Large Language Models	النماذج اللغوية الكبيرة
FLAIR	Fluid-Attenuated Inversion Recovery	تسلسل استعادة الانقلاب الموهن بالسائل
SVM	Support Vector Machines	آلات المتجهات الداعمة
GLCM	Gray-Level Co-occurrence Matrices	مصفوفات التواجد المشترك للمستوى الرمادي

Chapter 1: Introduction

Introduction

The last decade has witnessed a paradigm shift in Artificial Intelligence (AI) and Deep Learning (DL), revolutionizing the medical field, particularly radiology and neuro-oncology. Brain tumors, specifically gliomas, remain among the most aggressive malignancies, where early detection and precise histological grading are pivotal for treatment planning and patient survival. Medical imaging, particularly Magnetic Resonance Imaging (MRI), serves as the cornerstone of non-invasive diagnostics. However, relying on a single imaging modality often limits diagnostic performance, as distinct tissue characteristics are captured by different sequences [10], [12]. This project introduces a multimodal deep learning framework designed to merge features from multiple clinical sources, providing a comprehensive and accurate predictive system for brain tumor grading.

1.2 Statement of the Problem

Clinical diagnosis via computer-aided systems faces significant bottlenecks:

1. **Single-Modality Limitations:** A single MRI sequence cannot accurately and comprehensively differentiate between tumor cores, edema, and healthy tissue.
2. **Feature Complementarity:** Current systems often process sequences in isolation, failing to exploit the rich correlations across modalities.
3. **Clinical Reliability Deficits:** Models frequently suffer from data leakage and poorly calibrated probabilities, rendering them unsuitable for high-stakes medical decisions.

1.3 Research Objectives Primary Objective:

To build an intelligent, clinically robust multimodal ensemble framework for automated brain tumor grading. **Secondary Objectives:**

- Achieving a classification accuracy exceeding 90% with robust probabilistic calibration.
- Implementing a reproducible pipeline that eliminates data leakage.
- Providing a deployable API that supports flexible clinical decision-making.

1.4 Research Contributions

This thesis introduces several novel contributions:

- **Leakage-Controlled Ensemble Stacking:** Utilizing Out-of-Fold (OOF) stacking to ensure unbiased model generalization.
- **Clinical Calibration & Thresholding:** Deploying Platt Scaling [28] and dynamic decision thresholds to prioritize clinical safety (minimizing False Negatives).
- **Entropy-Driven Instance Selection:** Implementing Shannon Entropy-based slice selection for the Multiple Instance Learning (MIL) component [30] to filter volumetric noise.

1.5 Targeted Beneficiaries

- **Radiologists and Oncologists:** To assist in objective diagnostic stratification.
- **Medical Research Centers:** To provide a methodological foundation for predictive prognosis.
- **Medical AI Researchers:** As a reference for implementing strict calibration and leakage-prevention protocols.

1.6 Thesis Organization

The thesis is organized into six chapters: Chapter 2 details the theoretical mathematical and computational foundations. Chapter 3 presents the Systematic Literature Review (SLR). Chapter 4 elaborates on the methodological development and the proposed framework. Chapter 5 discusses experimental results and comparative analysis, and Chapter 6 concludes with final findings and future research prospects.

Chapter 2: Theoretical Background

2.1 Introduction

This chapter delineates the mathematical and computational foundations underpinning the proposed multimodal deep learning framework for glioma grading. The objective is to theoretically ground the concepts before delving into the applied methodology in subsequent chapters. This includes:

- 3D Convolutions and Residual Connections for volumetric representation.
- Vision Transformers and Self-Attention for long-range spatial dependency modeling.
- Multiple Instance Learning (MIL) and Shannon Entropy for informative slice selection.
- Ensemble Stacking, Out-of-Fold (OOF) predictions, and Logistic Regression meta-learning.
- Probability Calibration (Platt Scaling) and evaluation metrics (Brier Score, ECE).

2.2 Mathematical Foundations of 3D Convolutional Neural Networks

2.2.1 3D Convolution

In volumetric medical imaging, a volume is represented as a 4D tensor $x \in R^{C_{in} \times D \times H \times w}$. A 3D convolution kernel **K** is applied to produce a feature map **Y**:

$$Y_{c,d,h,w} = \sum_{c'=1}^{C_{in}} \sum_{i=0}^{k_D-1} \sum_{j=0}^{k_H-1} \sum_{k=0}^{k_W-1} K_{c,i,j,k} \cdot X_{c',\, d'+i,\, h'+j,\, w'+k} + b_c$$

Equation 2. 1:Mathematical formulation of the 3D Convolution operation

This operation ensures translation equivariance and enables early multi-modal fusion.

2.2.2 Residual Connections & ResNet Theory

Deep networks suffer from the vanishing gradient problem. Residual blocks solve this by learning the residual mapping $F(x)$alongside an identity shortcut:

$$x = F_{(x)} + x$$

Equation 2. 2:Mathematical formulation of a residual connection

The gradient flows directly through the addition operation:

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \frac{\partial \mathcal{F}}{\partial \mathbf{x}} + \mathbf{I}$$

Equation 2. 3:Derivative of the residual block mapping

The identity term **I** mitigates gradient vanishing, enabling the training of extremely deep networks like ResNet-50.

2.3 Vision Transformers & Self-Attention

2.3.1 Self-Attention Mechanism

For a sequence of tokens $\mathbf{z} \in \mathrm{R}^{N\times d}$, Query (**Q**), Key (**K**), and Value (**V**) matrices are computed. The attention is formulated as:

$$\mathrm{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}$$

Equation 2. 4 :Self-Attention calculation for token sequences

Scaling by $\sqrt{d_k}$ prevents the softmax gradients from saturating.

2.3.2 Shifted-Window Self-Attention (Swin)

To alleviate the quadratic $O(N^2)$complexity of global attention, Swin computes attention within local, non-overlapping windows of size **M**, reducing complexity to linear $O(N \cdot M^3)$. Shifting these windows across successive layers facilitates cross-window communication, making it highly effective for volumetric MRI classification [8], [21].

2.4 Multiple Instance Learning & Entropy Theory

2.4.1 Multiple Instance Learning (MIL)

In MIL, a patient is treated as a "bag" $\mathrm{B} = (\mathrm{x}_1 \cdot \cdots \mathrm{xn})$of slices (instances). A single bag-level label $Y \in \{0,1\}$ is assigned.

2.4.2 Gated Attention Mechanism

Instances are aggregated using gated attention weights [30] α_n:

$$\alpha_n = \frac{\exp(u_n \cdot g_n)}{\sum_{j=1}^{N} \exp(u_j \cdot g_j)}$$

Equation 2. 5 :Instance aggregation formulation using gated attention

where g_n acts as a learnable sigmoid gate that filters informative features prior to softmax normalization.

2.4.3 Shannon Entropy

Entropy is utilized to quantify the informativeness of an MRI slice s:

$$H(s) = -\sum_{b=1}^{B} p_b \log_2(p_b + \epsilon)$$

Equation 2. 6:Entropy-based formulation for MIL instance selection

Selecting slices with the highest entropy (Top-k) isolates diagnostically rich regions, effectively filtering out noise before the MIL aggregation.

2.5 Ensemble Learning & Meta-Learning

2.5.1 Out-of-Fold (OOF) Predictions

To avoid optimistic bias (data leakage) in stacking, base models generate Out-of-Fold predictions on held-out data. The meta-learner is exclusively trained on these OOF probabilities to ensure unbiased generalization.

2.5.2 Logistic Regression Meta-Learner

The meta-learner models the class probability using a Sigmoid function over a linear combination of base model probabilities:

$$P(\mathrm{HGG} \mid \mathbf{z}) = \sigma\left(\beta_0 + \beta_1 p_{\text{resnet}} + \beta_2 p_{\text{swin}} + \beta_3 p_{\text{mil}}\right)$$

Equation 2. 7:Class probability combination utilizing the meta-learner

2.6 Probability Calibration Theory

2.6.1 Platt Scaling

Deep neural networks are known to frequently output overconfident predictions. Platt Scaling rectifies this by applying a post-hoc logistic transformation [28] to the log-odds of the raw probabilities:

$$P_{\text{cal}} = \sigma(A \cdot \operatorname{logit}(\hat{p}) + B)$$

Equation 2. 8:Platt Scaling logistic mapping for raw probabilities

This monotonic transformation ensures the predicted probabilities align with the true statistical occurrences without altering the AUC-ROC ranking.

2.6.2 Evaluation Metrics

- **Brier Score:** Measures the mean squared difference between predicted probabilities and actual outcomes:

$$\text{Brier} = \frac{1}{N}\sum^{N}(\hat{p}_i - y_i)^2$$

Equation 2. 9:Brier Score formulation for probabilistic calibration evaluation

Expected Calibration Error (ECE): Quantifies the absolute difference between empirical accuracy and predicted confidence across multiple bins [29].

Chapter 3: Literature Review (Systematic Literature Review — SLR)

Introduction3.1

Previous studies play a pivotal role in framing any scientific research, enabling the researcher to understand the background of the problem, identify research gaps, and systematically build upon existing achievements. In the field of brain tumor classification and grading using Magnetic Resonance Imaging (MRI), recent years have witnessed remarkable progress—particularly with the integration of deep learning, hybrid architectures, multimodal learning, and ensemble learning—leading to diagnostically promising models in terms of accuracy. However, many of these studies still lack the necessary safeguards against data leakage and the probability calibration required for clinical adoption.

This chapter aims to systematically review and analyze the literature relevant to glioma grading (differentiating between High-Grade Gliomas (HGG) and Low-Grade Gliomas (LGG)) using deep learning. It focuses on model architectures, MRI sequences, preprocessing strategies, ensemble learning, and reliability metrics. The review statistically and analytically answers six Research Questions (RQs) based on 25 accepted studies documented in the review table, with the goal of demonstrating that the proposed multimodal deep learning framework in this project addresses genuine gaps in the literature.

3.2 Systematic Literature Review (SLR) Methodology

Search and Query Tool

This systematic review was conducted using the PARSIFAL tool, following a structured literature review methodology. The following query was utilized in the TITLE-ABS-KEY fields:

("Brain tumor" OR "Brain Cancer, Glioma, Brain Neoplasm" OR "MRI" OR "Magnetic Resonance Imaging, Neuroimaging") AND ("Deep learning" OR "Ensemble learning" OR "Multimodal learning" OR "Multi-modal, Data fusion, Hybrid learning") AND ("Model reliability" OR "Prediction accuracy" OR "Reliable Glioma grading (HGG vs LGG)" OR "Tumor classification" OR "Grading, Detection, Diagnosis")

Databases and Initial Results

The search was executed across the following databases, yielding these initial results:

Table 3. 1:Initial search results across selected digital databases

Database	Initial Result Count
IEEE	114
SCOPUS	803
ACM	457
NATURE	303
Approximate Total	**1676**

Screening and Acceptance Process

- **Title and Abstract Screening:** Inclusion and exclusion criteria were applied, resulting in an initial acceptance of 108 papers.
- **Full-Text Screening:** The criteria were reapplied to the 108 studies, l leaving 25 finally accepted studies. **The exhaustive data extraction for all accepted studies is documented in Appendix B.**

Inclusion Criteria (IC)

Table 3. 2:Inclusion Criteria (IC) applied for study selection

Code	Criterion
IC1	Focuses on grading/classifying brain tumors, specifically HGG vs. LGG, using clinical imaging (preferably MRI).
IC2	Relies on deep learning architectures or ensemble learning.
IC3	Reports standard performance metrics (e.g., Accuracy, F1-score, AUC).
IC4	Describes a clear and reproducible methodology and architecture.
IC5	Peer-reviewed papers (journals/conferences) published between 2018 and 2026.

Exclusion Criteria (EC)

Table 3. 3:Exclusion Criteria (EC) applied for study selection

Code	Criterion
EC1	Studies limited to detection or segmentation without a grading/classification component.
EC2	Vague methodology or incomplete evaluation results.
EC3	Reliance on non-imaging data without MRI.
EC4	Secondary research (systematic reviews, meta-analyses, editorials).
EC5	Traditional machine learning or non-AI-based techniques alone.
EC6	Non-human clinical data or animal models.
EC7	Duplicates or topics unrelated to brain tumors.

Research Questions (RQs) and their Statistical and Analytical Answers

The following answers are derived exclusively from the analysis of the 29 accepted papers.

RQ1: What is the distribution of research studies related to glioma grading using deep learning over time?

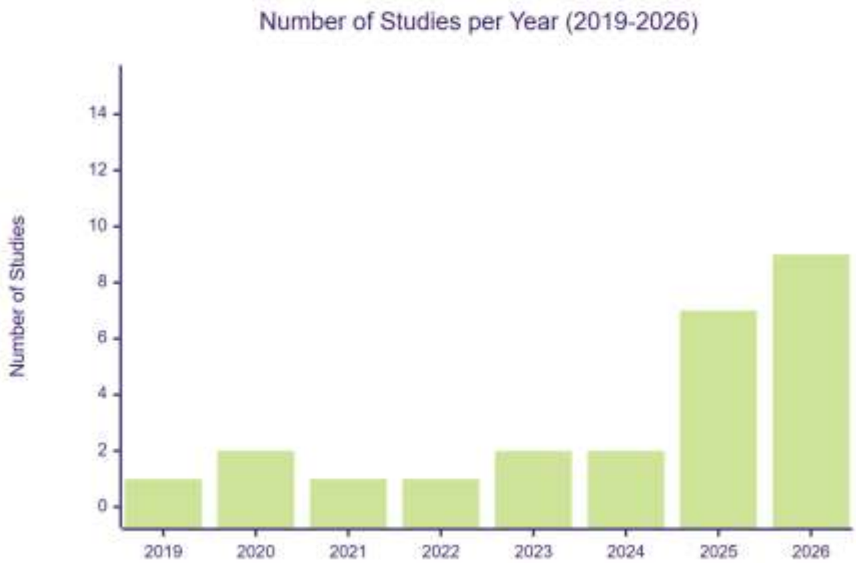

Figure 3. 1:Annual distribution of the 25 studies from 2019 to 2026
.

Analysis: There is a sharp chronological acceleration in publications. Between 2019–2024, the annual output remained modest, whereas 2025 and 2026 alone accounted for 16 studies (64.0%)—representing the vast majority of the sample. This is explained by the proliferation of Transformer/Mamba architectures, hybrid integration, and a growing interest in multi-center generalization. RQ1 confirms that the field is undergoing rapid technical maturation, necessitating rigorous evaluation methodologies (patient cross-validation, calibration) rather than merely settling for high accuracy.

RQ2: What are the primary data sources and MRI sequences used for glioma grading in the studies?

Data Sources (Explicitly mentioned in the 29 studies):

Table 3. 4:Primary data sources and datasets utilized in the reviewed studies.

Dataset / Source	Number of Studies Mentioning It
BraTS (2018–2021, 2020, etc.)	**21**
TCGA / TCIA	**~3 for each**

IXI (Healthy brain for comparison)	**2**
Private / Multi-center clinical data	**Several studies (e.g., Yin et al., Wu et al., Luo et al.)**

MRI Sequences:

Table 3. 5: Distribution of MRI sequences used across the selected studies.

Sequence	Number of Studies Using It
T1	23
T2	23
FLAIR	22
T1ce / T1CE	14
DWI	1

Input Modality Classification:

- **Multi-modality:** 22 studies (88.0%)
- **Single-modality:** 3 studies (12.0%) — e.g., T1ce only [Chatterjee et al., 2022], T2 only [Saluja et al., 2025], or DWI [Kaandorp et al., 2026].
- **Not clearly specified:** 1 study (3.4%)

Analysis: Multi-modal fusion has become the theoretical and practical standard (T1+T1ce+T2+FLAIR), aligning with clinical practice. However, 3 studies still rely on a single sequence, limiting generalizability. It is notable that BraTS serves as the experimental backbone (21/25), requiring caution against dataset bias without multi-center external validation.

RQ3: What are the common preprocessing techniques used to enhance MRI images for brain tumor classification?

Processing Level Classification:

Table 3. 6:Classification of preprocessing levels applied in the reviewed studies.

Category	Count	Percentage
Advanced	19	76.0%
Basic	6	24.0%

Recurring Techniques in Methodological Descriptions (Aggregated Qualitative Analysis):

- Co-registration and Skull-stripping (BET / HD-BET)

- Isotopic resampling to a standardized size (e.g., 1 mm³)
- Intensity normalization (Z-score / Min-Max / WhiteStripe)
- Cropping around the tumor (ROI) and simple resizing
- Data augmentation: Flipping, rotation, shifting
- Additional Advanced: Radiomics/IVIM extraction, histogram matching.

Analysis: The majority of works (76.0%) adopt an advanced processing pipeline that goes beyond simple resizing, reflecting an awareness of MRI challenges and the necessity for 3D volumetric preparation. The 24.0% relying on basic processing are predominantly associated with 2D slice-based methodologies, which inherently discard rich spatial geometry.

RQ4: How are deep learning architectures and ensemble strategies utilized to differentiate between HGG and LGG?

Methodological Classification (Simplified):

Table 3. 7:Methodological classification of utilized architectures (Simplified).

Architectural Style	Approximate Count	Percentage
Hybrid (DL + Radiomics/Attention)	18	72.0%
Transformer / Spatio-Temporal	4	16.0%
Ensemble Learning	2	8.0%
Standard DL	1	4.0%

Problem-Solving Patterns:

- **End-to-End Volumetric Classification:** Direct 3D classification using a shared backbone on a 4-channel tensor.2D slice-based classification (sometimes "best-slice").
- **Hybrid Segmentation-Grading Pipelines:** Segmentation strictly precedes classification to focus on ROIs.
- **Multi-task Learning:** Grading alongside molecular markers or survival prediction.
- **Ensembling:** Majority voting or stacking with a meta-learner.

Analysis: There is a clear shift toward hybrid architectures (72.0%) that merge local context (CNN) with auxiliary insights or global context. However, explicit ensemble learning appears in less than a tenth of the studies despite evidence of improved robustness. Furthermore, Multiple Instance Learning (MIL) is entirely absent in a rigorous mathematical form, representing a significant design gap.

RQ5: To what extent do current studies address data leakage and model calibration to ensure clinical reliability?

Leakage Control:

Table 3. 8:Status of data leakage control protocols.

Status	Count	Approximate Percentage
Strict patient-level isolation (High Reliability)	18	72.0%
Ambiguous, random, or leaky splits (Low Reliability)	7	28.0%

Probability Calibration:

Table 3. 9:Status of probability calibration reporting.

Status	Count
Uncalibrated / Not mentioned	21
Descriptive (Calibration curves / Brier)	4
Explicit Post-hoc Correction (e.g., Platt Scaling)	0

Reliability Category:

Table 3. 10:Classification of reviewed studies based on their overall clinical reliability.

Category	Count
High Reliability	18
Low Reliability	7

Analysis: Although 72.0% of studies enforce a strict patient-level split, 28.0% expose methodologies to inflated performance due to slice/case leakage. Clinically more critical is the complete absence (0%) of Platt calibration or strict post-hoc correction in the accepted literature. An overwhelming 84.0% report high performance indicators without linking them to reliable, calibrated probabilities, confirming that probabilistic clinical reliability remains a central design gap.

RQ6: What performance metrics are used to evaluate the models, and how are results reported for different glioma grades?

Metric Frequency (across 29 studies):

Table 3. 11:Frequency of evaluation metrics reported across the 25 studies.

Metric	Number of Reporting Studies
Accuracy	19
AUC / ROC-AUC	8
F1-score/ Macro-F1	9
Recall	6
Precision	4
Specificity	5
Sensitivity	2
Dice (mostly for segmentation)	5
C-index (survival)	1
Brier / ECE	Rare (2-3 mention Brier or curves)

Results Reporting by Grade:

- **Binary HGG/LGG:** The vast majority (traditional WHO grading task).
- **Multi-class / Molecular:** Wu et al. (IDH, 1p/19q, grade), Sun et al. (subtypes), Byeon et al. (GlioMT). Zhang et al. (multi-gene), Zeng et al. (5 subtypes).
- **Dual-task:** Segmentation + grading (Wen, Tripathi, SEMD-Net, Rajeev, Pei, etc.).

Analysis: Accuracy dominates as the primary metric across the 25 studies, despite the structural imbalance between HGG and LGG samples in medical practice. AUC is reported more frequently in recent robust studies to evaluate discriminative ability across thresholds. However, probabilistic calibration metrics (like Brier Score or Expected Calibration Error) are exceptionally rare. Furthermore, all reviewed studies assume a rigid, default 0.5 decision threshold, severely ignoring the clinical priority of minimizing False Negatives in HGG detection.

Introduction Conclusion: The answers to RQ1–6 prove that the field is moving toward multi-modality and 3D hybrid architectures, yet it technically lacks disciplined Out-of-Fold (OOF) ensembling, probabilistic calibration, and risk-aware thresholding. The following section details the most prominent studies, leading up to the definition of the design gap this research fills.

3 Previous Studies in Similar Domains.3

To maintain the narrative flow and readability of this chapter, the exhaustive Systematic Literature Review (SLR) matrix—detailing the architectures, datasets, evaluation metrics, and calibration status of all 25 accepted studies—is provided in **Appendix B**. The following section presents an in-depth analytical overview of 10 selected prominent studies that most closely align with the methodological scope of this research.

Below is an analytical overview of selected studies from the 29 reference papers, chosen based on recency, direct relevance to HGG/LGG grading, and architectural methodological diversity.

Study 1- Yin et al. [12]

- **Title:** Integrating deep learning and radiomics for preoperative glioma grading using multi-center MRI data
- **Proposal:** A multi-center framework integrating 3D deep learning with radiomics and Transformer attention for feature selection, incorporating attention-based feature fusion and Domain Adaptation for scanner variations.
- **Architecture:** 3D CNN backbone + Transformer attention module + RFECV for radiomics + feature fusion.
- **Metrics:** AUC, Brier, Calibration curves.
- **Strengths:** Use of multi-center data; acknowledgment of external validation; use of Brier/calibration in evaluation.
- **Methodological Weaknesses:** The design is retrospective; the dual architecture is computationally expensive; lacks an explicit ensemble model based on OOF Stacking to prevent implicit bias.

Study 2 — Wen et al. [4]

- **Title:** A deep ensemble learning framework for glioma segmentation and grading prediction
- **Proposal:** A multi-task ensemble deep learning framework based on an enhanced 3D U-Net model (GSG-UNet) connecting segmentation and grading.
- **Architecture:** Shared encoder with ACB blocks and Dual-Domain Attention + dual-branch decoder.
- **Metrics:** Accuracy, Recall, Specificity, Dice.
- **Strengths:** Functional integration between segmentation and grading; development of a dual-domain attention technique.
- **Methodological Weaknesses:** Fails to mention an explicit patient split protocol in segmentation, increasing the likelihood of data leakage; high sensitivity to missing imaging modalities; lacks Platt calibration processing.

Study 3 — Wu et al. [11]

- **Title:** Biologically interpretable multi-task deep learning pipeline predicts molecular alterations, grade, and prognosis in glioma patients

- **Proposal:** A Multi-Task Deep Learning (MDL) pipeline that predicts molecular markers, grade, and clinical prognosis with a focus on biological interpretability.
- **Architecture:** Shared 3D ResNet-10 encoder + multi-branch task heads.
- **Metrics:** AUC, C-index, calibration curves (for survival and prognosis).
- **Strengths:** Application of multi-institutional patient splitting; achieves structural clinical-molecular integration.
- **Methodological Weaknesses:** Relies partially on private data; structural complexity in interpreting joint multiple tasks; deviates from a qualitative focus on binary HGG/LGG classification.

Study 4 — Byeon et al. [21]

- **Title:** Interpretable multimodal transformer for prediction of molecular subtypes and grades in adult-type diffuse gliomas
- **Proposal:** GlioMT model — a multimodal transformer fusing MRI images with clinical text features encoded via a Large Language Model (BERT).
- **Architecture:** Transformer + BERT for clinical text; relies on attention maps for methodological interpretation.
- **Metrics:** AUC for multiple diagnostic tasks.
- **Strengths:** Introduces the first concrete integration of Large Language Models with radiological images in the context of gliomas.
- **Methodological Weaknesses:** Structural reliance on 2D slices loses spatial context; insufficient detail on strict patient splitting; lacks Platt Calibration mechanisms.

Study 5 — Li et al. [6]

- **Title:** MoRE-Net: An Interpretable and Modality-robust Model for Brain Tumor Grading
- **Proposal:** An interpretable model designed to be robust against missing imaging modalities, based on the Mamba architecture and a Prototypical Parts network.
- **Architecture:** MProtoNet + Mamba-based encoders + multimodal fusion with alignment loss.
- **Metrics:** Balanced Accuracy (BAC), Activation Precision (AP), Brier.
- **Strengths:** Innovative handling of the missing modalities challenge; adoption of the Brier score to assess reliability.
- **Methodological Weaknesses:** Overall discriminative power is considered moderate compared to full spatial architectures; requires complete modalities during the training phase; lacks an OOF ensemble strategy.

Study 6 — Ali et al. [5]

- **Title:** RGNN3D: A hybrid radiomic graph neural network for 3D MRI glioma grading
- **Proposal:** Fusion of manually extracted radiomic features with GCN+LSTM networks to represent patients as nodes in a graph, formulating a mechanism to prevent data leakage via an identity matrix.

- **Architecture:** PyRadiomics + GCN network + LSTM network; applies LIME-based interpretation.
- **Metrics:** Accuracy, F1, Precision, Recall.
- **Strengths:** Implementation of explicit patient isolation within the graph structure; use of LIME explanation.
- **Methodological Weaknesses:** Heavy reliance on manual geometric features; reduced model efficiency when relying on individual FLAIR sequences.

Study 7 — Gutta & Yathirajam [1]

- **Title:** Improving glioma grade classification with a hybrid CNN-transformer model
- **Proposal:** A lightweight hybrid model linking CNNs and the DeiT-Tiny model on 2D slices from the BraTS challenge.
- **Architecture:** CNN + Vision Transformer; extraction of interpretation maps via Grad-CAM.
- **Metrics:** F1, Accuracy, Precision, Recall.
- **Strengths:** Implementation of a 70/30 patient split; low temporal inference speed per slice; provides dual-structure interpretability.
- **Methodological Weaknesses:** Built on 2D dimensions while excluding essential T1 sequences; absence of a calibration step; limited to internal validation without exploring generalization.

Study 8 — Pei et al. [9]

- **Title:** Context aware deep learning for brain tumor segmentation, subtype classification, and overall survival prediction using radiology images
- **Proposal:** A spatially context-aware framework (CANet) to execute interconnected clinical tasks including segmentation, sub-classification, and survival prediction.
- **Architecture:** 3D CANet + 3D CNN + hybrid ML/DL architecture for prognosis prediction.
- **Metrics:** DSC for segmentation; overall ranking in standard challenges.
- **Strengths:** Functional synergy between multi-task integration; adoption of full-volume inputs.
- **Methodological Weaknesses:** High sensitivity to sample imbalance; the clinical grading pathway lacks a probability calibration mechanism.

Study 9 — Kobayashi et al. [19]

- **Title:** Observing deep radiomics for the classification of glioma grades
- **Proposal:** Introducing the concept of "Deep Radiomics" formulated via a CNN supported by Vector Quantization to ensure the stability of extracted features, followed by logistic regression.
- **Architecture:** Encoder-Decoder + VQ + Logistic Regression.
- **Metrics:** Accuracy, Specificity, Recall.
- **Strengths:** Methodological focus on structural interpretability and the numerical representation stability of features.

- **Methodological Weaknesses:** Lack of systematic external validation; relatively low Recall levels; reliance on a classical classification structure that diminishes the power of end-to-end deep modeling.

Study 10 — Saluja & Trivedi [24]

- **Title:** Glioma classification in MRI using a hybrid deep learning framework with majority vote ensemble
- **Proposal:** Ensembling five modified classical convolutional networks via Voting and Stacking strategies.
- **Architecture:** Fine-tuned CNNs + Majority/Weighted/Stacked Ensemble.
- **Metrics:** Accuracy, Sensitivity, Specificity, AUC.
- **Strengths:** Inclusion of an external validation protocol; exploiting architectural diversity in model ensembling.
- **Methodological Weaknesses:** Exclusive reliance on the T2 sequence alone; lack of clarity on the patient splitting protocol in the ensemble; absence of clinical probability calibration mechanisms; high risk of diagnostic slice data leakage.

3.4 Modern Techniques and Frameworks in the Field

Based on the systematic literature review (29 studies), the technical trends representing the State of the Art (SOTA) in glioma grading can be classified as follows:

3.4.1 Vision Transformers and Mamba

These architectures represent a leap in the ability to model long-range spatial dependencies compared to the local convolution of classical networks. Recent literature demonstrates significant efforts to integrate to represent complex contextual relationships within tumor tissues. The observed methodological challenge: Due to the high computational cost of full 3D volumes, most of these studies are forced to apply them to 2D slices or severely cropped volumes, causing the model to lose diagnostic spatial continuity.

3.4.2 3D Convolutional Neural Networks and ResNet

3D Convolutional Neural Networks (3D CNN) in their various branches, such as ResNet-based families, remain the most mature and practical choice for handling full volumetric spatial representations of MRI images. This is due to their efficiency in extracting local features and the ease of integrating Medical Transfer Learning techniques through them.

3.4.3 Ensemble Learning

Although theoretical reports prove that model fusion improves diagnostic robustness, disciplined ensemble learning applications remain limited in this field. The underlying technical gap: The Meta-learner is rarely designed to train exclusively on Out-of-Fold

(OOF) predictions. Most frameworks rely on direct voting integration or biased training that contributes to inflated apparent performance and decreased generalization.

3.4.4 Multiple Instance Learning (MIL)

MIL is theoretically the optimal choice for the tumor grading problem, where the medical label is available at the patient/full-volume level (the bag) while the diagnostic pathological evidence lies in specific slices (the instances). However, there is an almost complete absence of methodological designs that exploit this structural correlation in mathematically rigorous ways in the analyzed literature.

3.4.5 Graph Networks and Hybrid Models

Some recent trends focus on formulating frameworks based on Graph Convolutional Networks (GCN) or fusing manual Radiomics features with deep architectures. While they contribute to enhancing explanatory transparency, they increase the complexity of processing pipelines and limit the system's ability to learn directly from raw data without human intervention in feature formulation.

3.5 Comparative Analysis of Systems

When evaluating glioma grading systems, it is insufficient to solely compare Accuracy or absolute discriminative power. Multiple models may show high indicators while their diagnostic probabilities remain uncalibrated and are applied via a fixed, default decision threshold (0.5) that fails to account for clinical diagnostic priorities—especially minimizing False Negatives when mistakenly classifying a high-risk case (HGG) as low-risk (LGG), an error that could delay critical therapeutic intervention.

From this perspective, achieving clinical reliability requires integrating strong discriminative features with the provision of medically interpretable probabilities and flexible, adjustable operating points.

3.5.1 Multi-Axis Comparison Matrix

Table 3-1: Representative Comparison Between Reference Frameworks and the Proposed Methodological Design

Table 3. 12:Representative Comparison Between Reference Frameworks and the Proposed Methodological Design

Axis	Yin et al. [12]	Wen et al. [4]	Saluja & Trivedi [24]	Gutta & Yathirajam [1]	Proposed Methodological Framework (This Project)

Input Representation & Geometry	3D Multimodal + Radiomics	3D Multi-task U-Net	2D, T2 sequence only	2D slices, without T1	Full spatial 3D integrated with high-entropy slices MIL bag
Integrated Architectures	CNN + Transformer + Radiomics	U-Net + Attention	5 separate CNNs	CNN + DeiT-Tiny	Structural integration between: ResNet50-3D, SwinUNETR-3D, and DualStream MIL-3D
Ensemble Strategy	Feature fusion (not disciplined meta-inference)	Shared tasks (no meta-ensemble)	Traditional Stacking/Voting	Single hybrid model	Disciplined OOF Stacking with a Logistic Regression Meta-Learner
Data Splitting & Integrity Guarantee	Yes (multi-center split)	Not explicitly mentioned	Unclear and unknown	Yes (70/30 split)	Strict 5-Fold Stratified Patient-Level CV
OOF Predictions for Meta-Learner	No	No	Not mentioned	No	Yes — Exclusive evaluation on out-of-fold predictions to prevent performance inflation
Post-hoc Probability Calibration	No (used Brier for descriptive evaluation only)	No	No	No	Yes — Integration of Platt Scaling (independent split for calibration and threshold derivation)
Probability Quality (Brier / ECE)	Descriptive Brier index only	No mention	No mention	No mention	Targeted as a core design axis to minimize calibration error and

					improve confidence
Decision Threshold	Implicit and fixed at 0.5	Implicit and fixed at 0.5	Implicit and fixed at 0.5	Implicit and fixed at 0.5	Dynamic, clinically adjustable operating thresholds (Balanced / Sensitive to False Negatives)
Medical Risk Managem ent (FN for HGG)	Not included in methodology consideration s	Not structurally included	Not structurally included	Not structurall y included	Integrated by adjusting the operating threshold on a held-out set to minimize undetected cases

Table 3. 13:Probabilistic Clinical Reliability" Checklist in the Literature"

Methodological & Design Component	**Prevalence in the 25 Analyzed Studies**	**Formulation in the Proposed Framework**
Post-hoc Platt Scaling	**0 / 25 (0%)** of the published sample	Yes — Mathematical application on the log-odds of unbiased predictions.
Brier Score as an Optimization Axis	**4 / 25 (16.0%)** as marginal descriptive analysis	Yes — A targeted methodological standard to ensure the quality of probabilities.
Measuring Expected Calibration Error (ECE)	Very rare, almost non-existent	Yes — A core design metric to calculate confidence deviation across probability bins.
Liberation from the 0.5 Decision Threshold	Dominant and assumed by default	Yes — Derivation of custom thresholds serving medical diagnostic risk isolation.
Protecting Meta-learner from Training Data	Common overlap leading to inflated apparent accuracy	Yes — Restricting the meta-learner to strict OOF data to preserve neutrality.
Entropy-based MIL Context Selection	Rare and low adoption	Yes — A mathematical filter (Shannon) to exclude spatial noise.

Independent Protocol for Calibration & Thresholds	**0 / 25 (0%)** of previous studies	Yes — Strict structural separation of out-of-fold data.

Table 3. 14:Classical Analytical Comparison Matrix of Exploited Features and Methodological Gaps

Reference System / Study	Exploited Technical Features & Components	Key Strengths	Methodological Gaps (Compared to Current Design)
Yin et al. [12]	Fusion of 3D CNN and Transformer features with Domain Adaptation and manual features.	Robust external validation and reliance on multi-center data.	Lack of OOF-based Stacking strategy; clinically fixed decision threshold; omission of structural MIL mechanisms.
Wen et al. [4]	GSG-UNet architecture based on advanced convolution blocks to process joint segmentation and grading.	Functional synergy arising from the integration between segmentation and grading tasks.	Lack of strict safeguards against slice data leakage; omission of measuring and correcting confidence probabilities via Platt Calibration.
Saluja & Trivedi [24]	Constructing a post-hoc ensemble of five convolutional networks using classical Stacking.	Utilizing the architectural diversity of CNNs to improve discrimination.	Confined to the single T2 sequence; passing raw, overconfident probabilities through the default 0.5 threshold without calibration or ECE calculation.
Gutta & Yathirajam [1]	Formulating a hybrid model linking convolution blocks and the DeiT-Tiny model to process imaging slices.	High temporal inference speed and inclusion of Grad-CAM interpretation tools.	Reliance on 2D processing that loses volumetric context; absence of post-hoc probability calibration, raising doubts about its clinical reliability.
Proposed Methodology in this Research	Tri-architecture ensemble (Spatial, Transformers, and MIL) integrated with Platt calibration and	Strict structural integration between robust discrimination and flexible, clinically adjustable	Commitment to building and tuning on the scale of available volumetric data for development and validation (discussed

	flexible operating thresholds derived out-of-fold.	probabilistic calibration.	in detail within methodology limitations in subsequent chapters).

3.5.2 Axis 1: Probability Calibration (Platt Scaling) vs. "The Illusion of Discriminative Accuracy"

The analysis of the **25 reference studies** reveals that exactly **84.0%** of them completely lack any component related to the post-hoc calibration of diagnostic probabilities, with **zero studies (0/25)** explicitly and systematically applying a Platt Scaling protocol to correct probability confidence values for glioma cases. While previous works focus on highlighting high accuracy magnitudes, these indicators merely reflect the model's quality in terms of Discrimination, but completely fail to describe the reliability of the numerical probability itself, which is known as Calibration.

The Clinical Problem of Overconfidence:

When an uncalibrated medical model outputs a confidence value such as P(HGG)=0.90, this number does not necessarily correspond to the diagnostic statistical reality. This deviation could guide an oncologist toward adopting excessively aggressive surgical protocols, or vice versa, based on a geometrically misleading probabilistic indicator.

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To bridge this common structural gap in the literature, the proposed methodological design in this research incorporates a Post-hoc Platt Scaling step as a fundamental structural requirement applied directly to the log-odds of the unbiased predictions extracted from the ensemble models via the OOF mechanism:

$$P_{\text{cal}} = \sigma\left(A \cdot \log \frac{P_{\text{uncal}}}{1 - P_{\text{uncal}}} + B\right)$$

Equation 3. 1:Platt Scaling transformation for calibrating model probabilities

Where the parametric coefficients (A and B) are tuned to minimize probabilistic deviation. This clinical robustness is measured by calculating the Brier Score (the direct squared error between the probability and the true label) and the Expected Calibration Error (ECE) index, which represents the expected value of calibration error by dividing the probability range into uniform bins:

$$\text{ECE} = \sum^{B} \frac{n_b}{N} \left|\text{acc}(b) - \text{conf}(b)\right|$$

Equation 3. 2:Calculation of Expected Calibration Error across probability bins

This structural construction in our current model aims to shift the focus from merely achieving abstract discriminative power (represented by the overall ensemble AUC of the model) to subjecting the resulting probabilities to a mathematical optimization process that guarantees a tangible reduction in Brier Score and ECE values. This design approach ensures that the system's probabilistic outputs become a tool of genuine confidence reflecting the actual distribution of diseases within clinical settings, representing a fundamental methodological shift that separates the current framework from the vast majority of the reviewed literature that settles for presenting classical accuracy metrics.

3.5.3 Axis 2: Dynamic Threshold Strategy as an Alternative to the Rigidity of the Default 0.5 Threshold

All analyzed reference studies—in a manner that ensures automatic repetition in design rely on a fixed, rigid decision threshold of 0.5 to determine the final classification outputs, without providing any medical or clinical justification for this choice. In brain tumor diagnosis environments, this rigidity represents a methodological hazard, given that the danger of False Negatives (missing an HGG tumor and classifying it as LGG) far exceeds the consequences of False Positives. This makes choosing a fixed threshold an incompatible option with medical risk management principles.

The Proposed Methodological Innovation in the Decision Plan: To compensate for this shortcoming, the current design introduces a flexible methodology to derive Dynamic Operating Points extracted through a disciplined division of the Out-of-Fold (OOF) prediction data. A specific, completely independent portion is isolated for calibration purposes, and another held-out portion is exclusively dedicated to deriving decision policies and optimizing thresholds, preventing circular evaluation or data leakage into the parametric tuning phase.

This methodology includes formulating distinct decision policies that cater to real clinical scenarios via a systematic Threshold Sweep protocol:

- **Balanced Decision Policy($t_{balanced}$):** An operating threshold derived to achieve the optimal, maximum balance between Precision and Recall, targeting general medical applications that require parity in handling diagnostic error from both sides.

- **High Sensitivity Policy for Minimizing False Negatives ($t_{sensetive}$):** A shifted, clinically optimized decision threshold that gives top priority to raising Recall rates and detecting critical HGG cases, even at the expense of a slight, medically acceptable increase in False Positive rates, serving high-risk clinical screening and triage pathways.

Integrating the mechanism for deriving operating thresholds based on the sorting and analysis of probabilities *only after* they have been calibrated grants the system an applied flexibility that all previous studies lack, as they treat the 0.5 threshold as a rigid mathematical law immune to adaptation in the interest of patient safety and physician decision-making.

3.5.4 Axis 3: Multiple Instance Learning (MIL) Based on Shannon Entropy (Top-k Information Context)

Although the volumetric nature of MRI images perfectly aligns with the hypotheses of Multiple Instance Learning (MIL), the literature review reveals a severe scarcity in formulating frameworks that exploit this concept. Most previous models settle for either processing the entire volume along with its spatial noise, or resorting to manual, selective choice of specific slices based on the researcher's vision, which weakens the system's automated generalization mechanism.

The methodological design of DualStreamMIL-3D in this project acts as a direct response to bridge this literary gap by introducing a disciplined mathematical criterion to frame the modeling inputs based on Information Shannon Entropy. This design innovation is summarized in the following steps:

First, the Shannon Entropy Value H(s) is calculated for each axial slice within the medical volume, based on the statistical distribution of diagnostic signal intensities after subjecting them to standard normalization processes, while including a numerical protection factor (ε) to ensure computational stability:

$$H(s) = -\sum_{i=1}^{256} p_i \log_2(p_i + \epsilon)$$

Equation 3. 3:Modified Shannon Entropy formula for assessing MRI slice informativeness

Second, all spatial slices are arranged in descending order according to their informational content. Subsequently, a strict systematic filter is applied to select the best set of slices richest in structural data and histological details (Top-k most informative), automatically excluding background slices or peripheral sections that represent computational noise distracting the efficiency of deep models.

Third, the "MIL Bag" is constructed based on these selected slices to feed the multiple instance learning pathways, ensuring the integration of localized spatial context while focusing on critical regions within the tumor formation. This design approach contributes to filtering inputs and protecting the model from unhelpful dimensional inflation, thereby elevating the quality of diagnostic feature extraction and reducing clinical error rates compared to traditional methodologies that treat spatial volumes as a single block without informational differentiation. These structural choices are subject to strict experimental evaluation and detailed architectural comparisons that are quantitatively reviewed to confirm the efficiency of the methodological design in the results and subsequent evaluation chapter.

Chapter 4: Research Methodology

4.1 Introduction

This chapter details the methodological framework developed to address the brain tumor grading problem, specifically the binary classification of High-Grade Glioma (HGG) versus Low-Grade Glioma (LGG). Section **4.5** outlines the dataset and preprocessing protocols, while Section **4.2** provides a systematic analysis of traditional approaches. Section **4.3** elaborates on the proposed ensemble framework, which integrates three complementary architectures: a 3D ResNet50, a transformer-based SwinUNETR-3D, and a DualStreamMIL-3D model based on Multiple Instance Learning. Furthermore, we describe the mechanisms for aggregating predictions via meta-learning, post-hoc calibration, and the establishment of clinical decision thresholds.

This chapter serves as the cornerstone of the thesis regarding the **methodological framework**, delineating the technical mechanisms constructed under a rigorous paradigm to address critical clinical application challenges: **data leakage**, **class imbalance**, **unreliable raw probability estimation**, and the necessity for **tunable operating points** to balance sensitivity and specificity. The methodology is grounded in a comprehensive architectural analysis and a reliable evaluation framework for leakage-controlled glioma grading (*Reliable Glioma Grading via Leakage-Controlled*).

Organizational Note: This chapter focuses exclusively on **methodological mechanisms and procedures**. Quantitative results and experimental comparisons—including performance metrics, finalized decision thresholds, and error analyses—are comprehensively discussed in **Chapter 5: Results and Discussion**.

4.2 Traditional Methods for the Problem

4.2.1 Numerical Methods

Traditionally, quantitative analysis of brain tumors has relied on numerical and statistical methods to extract features from MRI scans. Such metrics include tumor volume, enhancement ratios, and signal intensity indices across various sequences (T1, T1ce, T2, and FLAIR). Furthermore, descriptive and inferential statistical analyses—such as computing skewness and kurtosis of intensity distributions within the Region of Interest (ROI) or employing hypothesis tests (e.g., t-tests)—have been widely utilized to characterize differences between tumor grades. Despite their interpretability, these methods face fundamental limitations: (1) an over-reliance on hand-crafted feature engineering which necessitates extensive radiological expertise [22], [26]; (2) high sensitivity to inter-scanner and inter-protocol variations; (3) an inability to capture complex 3D spatial relationships; and (4) limited generalization capabilities when faced with the diverse morphological variations of gliomas. These constraints have necessitated the transition toward machine learning and, subsequently, deep learning approaches.

4.2.2 Machine Learning Methods

Traditional machine learning approaches have utilized algorithms such as Support Vector Machines (SVM), Random Forests, and Logistic Regression, applied to features extracted from MRI scans—ranging from statistical descriptors to texture analysis using Gray-Level Co-occurrence Matrices (GLCM) or wavelet-based features. While traditional machine learning models offer clarity and lower computational requirements, they often encounter a bottleneck in the efficacy of manual feature engineering. Specifically, these models struggle with: (1) interactive multi-modal data fusion; (2) modeling long-range spatial dependencies within 3D brain volumes; and (3) accommodating the significant intra-class variability inherent in HGG and LGG cases. Consequently, this research adopts a deep learning approach—specifically, a multimodal multi-architecture ensemble framework—to automatically learn hierarchical representations, thereby effectively addressing the complex clinical challenges that traditional methods fail to resolve.

4.3 Model Architecture and Training Strategy

4.3.1 Overview of the Proposed Framework

This research proposes an **ensemble framework** that integrates predictions from three base models with complementary **inductive biases**, allowing the system to leverage the unique strengths of each architecture in addressing brain tumor classification challenges:

.

Table 4. 1:Methodological roles and architectural types of the base models in the proposed ensemble framework

Model	Architecture Type	Methodological Role
ResNet50-3D	3D CNN	Hierarchical local spatial feature extraction across the full volume.
SwinUNETR-3D	3D Transformer	Modeling long-range spatial dependencies and global context.
DualStreamMIL-3D	Multiple Instance Learning (MIL)	Slice-level inference identifying highly diagnostic instances.

HGG probability scores are aggregated using a **logistic regression meta-learner**, followed by **post-hoc Platt scaling** for probability reliability, and finally, **clinical operating point policies** defined on an independent hold-out set.

4.3.2 Base Models

A) ResNet50-3D (Volumetric Baseline)

The **ResNet50-3D** (MedicalNet-compatible) acts as the CNN baseline. It accepts a 4-channel early-fused 3D volume (B, 4, 128, 128, 128) and comprises an initial 3D convolution, four stages of Bottleneck3D blocks (depths= [3, 4, 6, 3]), and a classification head. By loading optional MedicalNet weights with a 4-channel conv1 adaptation, this model serves as a representative of **convolutional inductive bias**, focusing on efficient local hierarchical feature extraction throughout the entire brain volume.

B) DualStreamMIL-3D (Localized Reasoning)

Adopting a **Multiple Instance Learning (MIL)** paradigm, this model treats each patient as a **bag** of 16 axial slices selected via an **entropy-based top-k algorithm**. The architecture features:

1. **Instance Encoder:** A 4-channel-adapted ResNet18 for slice feature extraction.
2. **Dual-Stream Design:** Integrating a "Critical Instance Selector" (MLP + temperature-scaled Softmax) and a "Contextual Aggregator" (Gated Attention). This approach enables inference at the slice level without discarding patient-level labels, complementing volumetric models by isolating the most informative diagnostic evidence.

C) SwinUNETR-3D (Global Context Modeling - Extensive Breakdown)

SwinUNETR-3D serves as the core of the framework, leveraging the **Swin Transformer encoder (Swin ViT)** from MONAI, adapted with a direct classification head.

- **Transformer Motivation:** Unlike CNNs, which have limited receptive fields, **Self-Attention** allows the model to relate arbitrary voxels. This is crucial for brain MRI, where HGG indicators (e.g., heterogeneity, contrast enhancement, peri-tumoral effects) are often spatially distributed.
- **Patch Embedding:** The (B, 4, 128, 128, 128) input is partitioned into 3D patches, linearly embedded into a token dimension of 48. This enables a balance between spatial resolution and computational feasibility.

- **Hierarchical Design:** The model employs four stages (depths= [2, 2, 2, 2]) with increasing attention heads ([3, 6, 12, 24]). The feature dimension expands (48 → 96 → 192 → 384), reaching a final representation depth of **768**.
- **Shifted-Window Self-Attention:** This mechanism computes attention within local 7×7×7 windows. Alternating shifted-window blocks facilitate cross-window communication, enabling the model to capture **global representations** with local computational costs.
- **Classification Head:** Features are processed via **Global Average Pooling** followed by a Linear (768 → 2) layer. Differential learning rates (5×10^{-5} for the encoder, 1×10^{-4} for the head) are applied to preserve pretrained feature integrity while fine-tuning the classification layer.
- **Strategic Role:** SwinUNETR-3D acts as the primary driver of the ensemble, providing robust, context-aware representations that effectively identify subtle indicators of tumor grade.

4.3.3 Loss Function and Training Algorithms

Loss Function All base models are trained using **CrossEntropyLoss** for binary classification (LGG=0, HGG=1), defined as:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{p}_i) + (1-y_i)\log(1-\hat{p}_i)\right]$$

Equation 4. 1:Objective loss function formulation for binary glioma classification

Rationale: This function provides stable gradients, facilitates seamless integration with WeightedRandomSampler for class imbalance handling at the data level, and ensures a standardized objective function for cross-model comparability.

Optimizers Models utilize **AdamW** or **Adam** with parameter-grouping for the backbone and classification head. Key hyperparameters include:

- Weight Decay: 1×10^{-4}.
- Batch Size: 4 (for 3D models).
- Max Epochs: 60.

Learning Rate Scheduling (Cosine Annealing with Warmup) A structured learning rate schedule is employed:

$$\eta_t = \begin{cases} \eta_{max} \cdot \frac{t+1}{T_{warmup}} & t < T_{warmup} \\ \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min}) \left(1 + \cos\left(\pi \cdot \frac{t - T_{warmup}}{T - T_{warmup}}\right)\right) & t \geq T_{warmup} \end{cases}$$

Equation 4. 2:Mathematical formulation of the learning rate scheduler

Where (T warmup) denotes the initial ramp-up phase, ensuring stable convergence early in the training process.

Early Stopping

To mitigate overfitting, training is monitored via **Validation AUC**. Patience is set to 10 epochs (min epochs: 10), with F1-score acting as a tie-breaker. The best-performing checkpoint (best.pt) is preserved per fold.

4.3.4 Post-hoc Calibration, Ensemble Integration, and Decision Thresholding

Methodological Philosophy: Clinical reliability requires not only high discriminative performance but also **well-calibrated probabilities**. This framework employs an **OOF Stacking** strategy combined with **Platt Scaling** and a strictly partitioned data split (70/30) to eliminate **data leakage**.

(1) Out-of-Fold (OOF) Predictions & Leakage Prevention

To prevent optimistic bias, the meta-learner is trained exclusively on OOF predictions, where each prediction is generated by a model on data it has not encountered during its respective training fold.

Fold k: Train on {0...4} \ {k} → Predict on fold k (OOF_k)

OOF = concat (OOF_0, ..., OOF_4)

This ensures unbiased patient-level evaluation.

(2) Meta-Learner (Logistic Regression)

Instead of simple averaging, a weighted linear combination is learned to maximize discrimination between HGG and LGG.

$$P_{\text{uncal}}(\text{HGG}) = \sigma\left(\beta_0 + \sum_{j=1}^{3} \beta_j \cdot p_j\right)$$

Class weighting is set to balance to account for the HGG/LGG distribution.

Equation 4. 3:Uncalibrated ensemble probability aggregation formulation

(3) Post-hoc Platt Scaling Calibration is performed on the log-odds of the meta-learner output to ensure output probabilities correspond to clinical frequencies:

$$P_{\mathrm{cal}} = \sigma(A \cdot \mathrm{logit}(P_{\mathrm{uncal}}) + B)$$

Equation 4. 4:Application of Platt Scaling to the ensemble predictions

This step is performed **post-hoc** (after base training), ensuring the model's ranking ability remains intact while improving reliability.

(4) Data Partitioning (70/30 Split)

- **Calibration Set (70%):** Used to fit the Platt calibrator.
- **Threshold Selection Set (30%):** Used to sweep thresholds τ and select clinical operating points.

  This strict separation prevents **Circular Leakage** and ensures that threshold performance metrics are indicative of true generalization.

(5) Threshold Tuning Policies

Thresholds (tau(t)) are tuned on the independent 30% hold-out set:

1. **Balanced (Max F1):** Optimized for general screening.
2. **High-Sensitivity:** Optimized to minimize False Negatives (FN), essential for high-risk clinical triage where missing an HGG case carries severe consequences.

(6) Nested CV

The framework employs a **nested cross-validation** protocol for the meta-learner to confirm that the calibration and thresholding strategies are robust across different data partitions, ensuring reproducibility.

4.4 Detailed Pipeline Explanation

The project methodology is structured into nine sequential stages, ensuring rigorous data integrity from raw acquisition to final clinical inference. The following diagram illustrates the operational workflow:

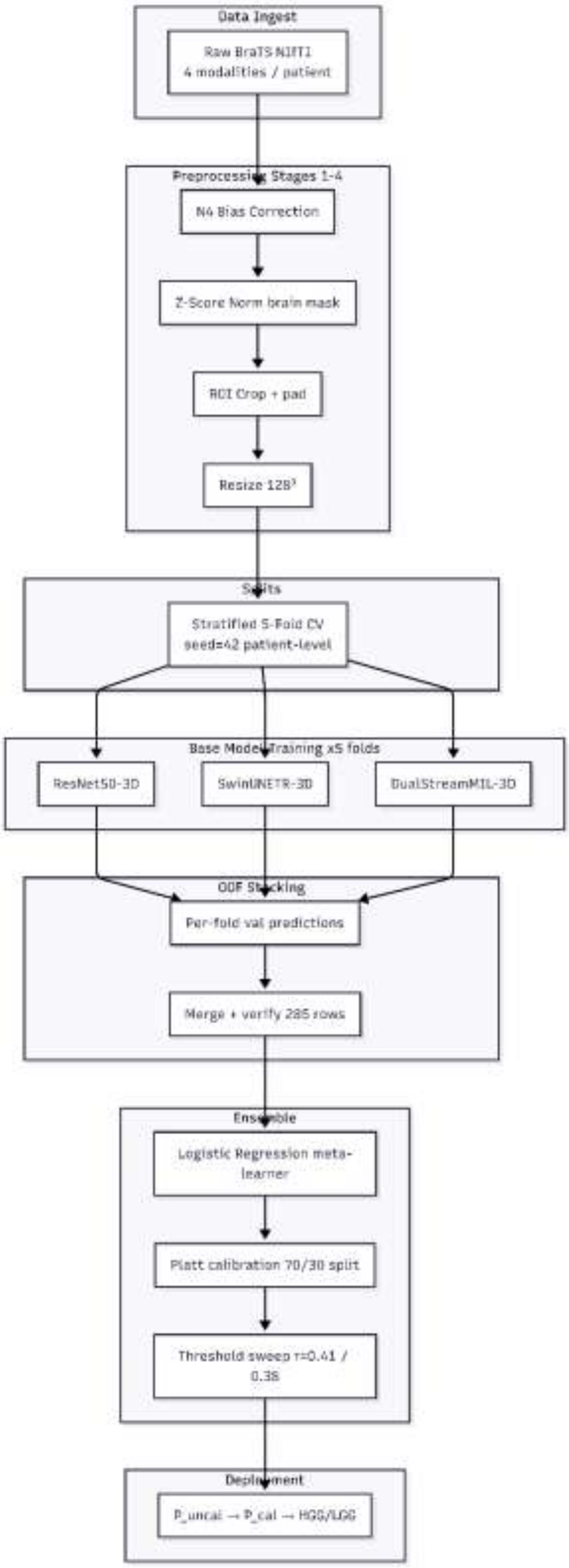

Figure 4. 1:Proposed Pipeline Architecture for Leakage-Controlled Glioma Grading

Table 4. 2:End-to-End Execution Pipeline of the Proposed Framework

Stage	Process	Key Inputs	Key Outputs
.1Data Ingest	Acquisition of raw MRI modalities	Raw NIfTI (4 modalities)	Structured subject directories
.2Preprocessing	Stages 1-4: Intensity & Spatial normalization	Raw MRI	Standardized 128^3 volumes
.3Split Generation	Patient-level stratified CV (5-fold)	Subject indices	-5fold CSV indices
.4Base Training	Training 3 distinct architectures	Standardized volumes	Model checkpoints
.5OOF Generation	Inference on held-out folds	Trained model folds	Per-model OOF CSVs
.6Data Merging	Concatenation and integrity verification	OOF CSV files	Unified OOF table (285 patients)
.7Meta-Learner	Logistic Regression ensemble training	OOF probability vectors	Meta-learner model
.8Calibration	Platt scaling and operating point selection	Meta-learner outputs	Calibrated models & thresholds
.9Deployment	Clinical inference via FastAPI	New subject 4×NIfTI	Calibrated HGG/LGG decision

4.5 Dataset and Preprocessing Pipeline

4.5.1 BraTS 2018 Dataset

The Brain Tumor Segmentation Challenge (BraTS) 2018 dataset [25] served as the foundational cohort for training and evaluating the proposed framework.

Table 4. 3:Key attributes and specifications of the BraTS 2018 dataset used in this study.

Attribute	Value
Source	BraTS 2018 (MICCAI Challenge)
Cohort Size	285patients (210 HGG, 75 LGG)
Class Imbalance	2.8:1(1HGG:LGG)
Modalities	T1, T1ce (Contrast-Enhanced), T2, FLAIR
Labeling	Patient-level binary classification (HGG=1, LGG=0)
Split Index	splits/kfold_5fold_seed42.json

4.5.2 Disk-Based Preprocessing (Stages 1–4)

A four-stage preprocessing pipeline was strictly executed to achieve **Data Standardization**. This ensures consistency in intensity distributions, spatial dimensions, and anatomical focus across the cohort, providing a stable learning environment: Raw NIfTI → Stage 1 → Stage 2 → Stage 3 → Stage 4 → Training-Ready Volumes

Table 4. 4:Detailed specifications of the MRI preprocessing pipeline stages

Stage	Objective and Procedure	Technical Details
1. N4 Bias Correction	Mitigate scanner-induced intensity inhomogeneity.	Employed N4BiasFieldCorrection with an Otsu mask. Iterations: [40,40,30,20].
2.Z-Score Normalization	Standardize statistical distributions of pixel intensities.	Applied$\frac{(x-\mu)}{(\sigma+\varepsilon)}$ exclusively to brain voxels($x = 10^8$)
3. ROI Cropping	Reduce computational overhead by removing background.	Bounding box extraction based on the FLAIR modality, incorporating a 10-voxel padding.
4. Resampling	Unify spatial dimensions for batch processing.	Linear interpolation to an isotropic volume of **128×128×128**.

4.5.3 Runtime Data Augmentation

To enhance generalization and prevent overfitting, stochastic geometric transformations were dynamically applied via the MONAI library exclusively during **training**:

- Random Rotation (±15° per spatial axis).
- Random Flip (50% probability per axis).
- Random Zoom (±10% scaling).
- Random Translation (±10% of the volumetric extent).

4.5.4 Class Imbalance Mitigation

To counteract the dataset's inherent bias toward the HGG class, a WeightedRandomSampler was utilized to dynamically oversample the minority LGG class during training. Sample weights were computed using inverse class frequency:

$$w_c = \frac{N_{total}}{N_{classes} \times N_c}$$

Equation 4. 5:Sample weight formulation for the WeightedRandomSampler

(LGG weight: 1.90; HGG weight: 0.68).

4.5.5 Data Partitioning Strategy

To guarantee an unbiased, leakage-free evaluation, a **Patient-Level 5-Fold Stratified Cross-Validation** protocol was implemented. This strictly ensures that no slices or volumes belonging to the same patient co-occur in both the training and validation sets.

4.5.6 Information-Theoretic MIL Slice Selection

The DualStreamMIL-3D model identifies the most diagnostically informative slices rather than relying on random selection:

1. Computation of **Shannon Entropy** for each axial slice (using a 256-bin histogram on normalized intensities).
2. Sorting and extraction of the most informative **Top-16 slices**.
3. Aligning the selected slice indices across all four modalities to construct a cohesive multi-modal MIL bag.

4.6 Chapter Summary and Methodological Contributions

4.6.1 Chapter Summary

This chapter delineated a comprehensive research methodology for glioma grade classification by architecting an integrated multimodal ensemble framework. The pipeline strategically fuses the inductive biases of three architectures: (1) ResNet50-3D for local volumetric patterns, (2) SwinUNETR-3D for global contextual dependencies, and (3) DualStreamMIL-3D for localized slice-level reasoning. To clinically contextualize this integration, a meta-learner was trained on Out-of-Fold (OOF) predictions, followed by post-hoc Platt scaling and the definition of tunable operating thresholds.

4.6.2 Core Methodological Contributions

Table 4. 5:Summary of the research contributions and their methodological justifications within the proposed framework.

#	Contribution	Methodological Justification
1	Strict Leakage Control	Enforcement of OOF stacking, patient-level splits, and a rigorous 70/30 segregation for calibration and thresholding.
2	Hybrid Architecture Integration	Synergizing CNNs, Transformers, and MIL paradigms to overcome single-architecture limitations.

3	Clinical Probability Reliability	Implementation of Post-hoc Platt Scaling to align predicted probabilities with actual statistical occurrences.
4	Dynamic Clinical Controllability	Formulation of dual threshold policies (Balanced and High-Sensitivity) allowing clinicians to manage the FP/FN trade-off.
5	Entropy-Driven Instance Extraction	Utilizing information theory to filter out noisy background slices, thereby refining the MIL input bag.

4.6.3 Conclusion and Transition to Chapter 5

This chapter marks the transition from conceptualizing an isolated classification model to engineering a **deployable, clinically reliable system** where predictions are treated as interpretable medical decisions. This robust methodological foundation sets the stage for **Chapter 5 (Results and Discussion)**, which will present the quantitative and empirical validation of these mechanisms. The upcoming chapter will detail discriminative performance metrics, calibration efficacy (Brier and ECE scores), and the empirical impact of the established clinical thresholds.

Chapter 5: Results and Discussion

Introduction This chapter presents and analyzes the experimental results achieved by the proposed multimodal deep learning framework for glioma grading (HGG vs. LGG). The system was evaluated using a rigorous 5-Fold Stratified Patient-level Cross-Validation protocol on the BraTS 2018 dataset (285 patients). This chapter details the quantitative performance of individual models, the ensemble meta-learner, and the clinical evaluation based on probability calibration and dynamic operating thresholds. Finally, a critical comparison is drawn between the proposed framework and State-of-the-Art (SOTA) models identified in the Systematic Literature Review.

5-1 Tools and Evaluation Metrics

To ensure a comprehensive evaluation encompassing both discriminative power and clinical reliability, two sets of metrics were employed:

1. Discriminative Metrics: Standard binary classification metrics:

- **Accuracy:** The overall percentage of correct predictions.
- **Recall / Sensitivity:** A clinically critical metric to minimize False Negatives (FN):

$$\text{Recall} = \frac{TP}{TP + FN}$$

Equation 5. 1:Calculation of the Recall metric to assess False Negative minimization

Precision:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Equation 5. 2:Calculation of Precision for diagnostic prediction

- **F1-Score:** The harmonic mean of Precision and Recall.
- **AUC-ROC:** Evaluates the model's ability to rank cases correctly independent of the decision threshold.

2. Reliability and Calibration Metrics:

In critical medical contexts, it is imperative to evaluate how well model probabilities align with actual statistical frequencies (avoiding overconfidence):

- **Brier Score:** Measures the mean squared error between the predicted probability and the true label (closer to 0 is better).

- **Expected Calibration Error (ECE):** Calculates the average gap between empirical accuracy and predicted confidence across 10 probability bins:

$$\text{ECE} = \sum_{b=1}^{B} \frac{n_b}{N} \left|\text{acc}(b) - \text{conf}(b)\right|$$

Equation 5. 3:Expected Calibration Error formulation across 10 probability bins

These metrics are highly suitable for the problem at hand, as relying solely on AUC may mask overconfident probabilities that lead to misleading clinical decisions.

5-2 Proposed Model Results (The Reliable Ensemble Framework)

The Out-of-Fold (OOF) Stacking ensemble framework represents the final core output of this research, demonstrating high discriminative ability coupled with exceptional clinical reliability.

Meta-Learner Performance:

The Logistic Regression meta-learner, trained on the OOF predictions of 285 patients, assigned the following coefficients:

- `SwinUNETR-3D`: **+4.06** (Dominant contributor capturing global context).
- `DualStreamMIL-3D`: **+0.89** (Important complementary signal for high-entropy slices).
- `ResNet50-3D`: **+0.54** (Secondary contributor for local volumetric features).

Full Dataset Performance (Full OOF, n=285)

The ensemble framework achieved an **AUC-ROC of 0.9126** and a PR-AUC of **0.9705**.

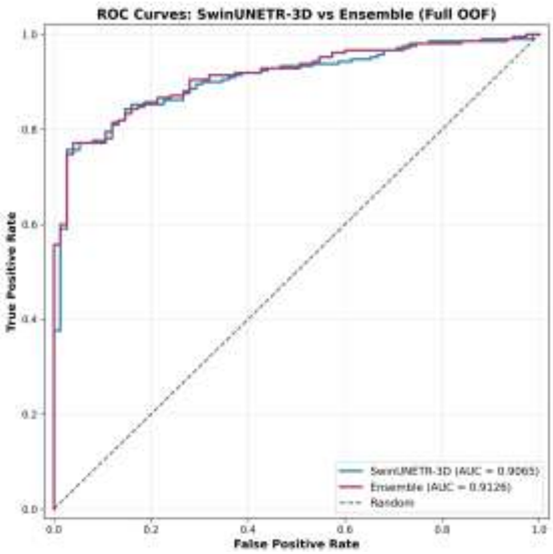

Figure 5. 1:ROC curves comparing the performance of the SwinUNETR-3D model and the proposed ensemble pipeline

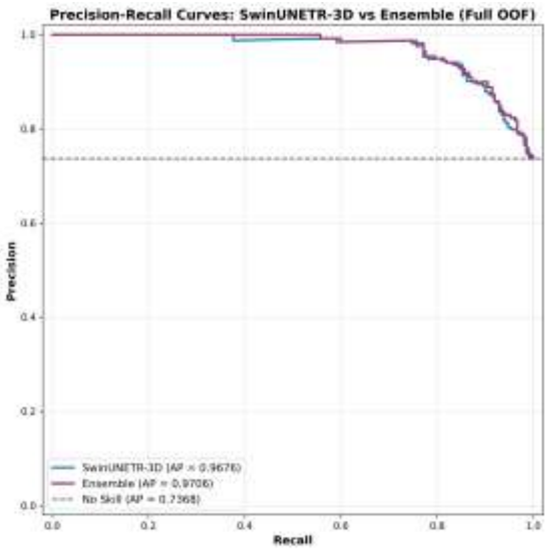

Figure 5. 2:Precision-Recall curves comparing the SwinUNETR-3D model and the proposed ensemble pipeline

Calibration Results and Clinical Operating Thresholds:

The Platt scaling technique successfully achieved its goal. On the held-out threshold selection set (86 patients), the Brier Score was reduced from 0.119 to **0.099** (a **16.8%** improvement), and ECE dropped from 0.119 to **0.087** (a **26.9%** improvement).

Based on calibrated probabilities, two clinical operating points were derived:

1. **Balanced Threshold ($\tau = 0.41$):** Achieved an F1-score of 0.9365 with perfect error balance (FN=4, FP=4).
2. **High-Sensitivity Threshold ($\tau = 0.38$):** Achieved high Recall (0.9524), minimizing False Negatives to just **(FN=3)**. This is the recommended threshold for high-risk clinical triage.

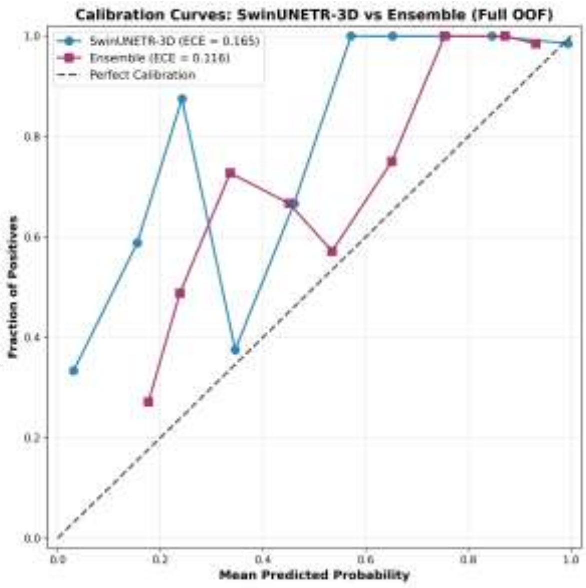

Figure 5. 3:Calibration curves on full OOF predictions (n = 285); ECE reported in the legend.

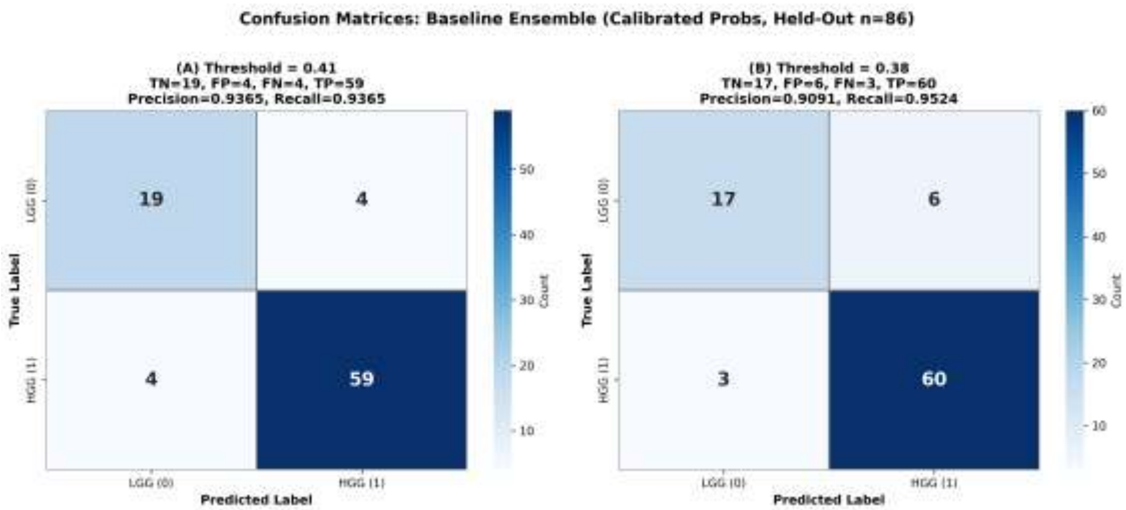

Figure 5. 4:Confusion matrices on the held-out threshold-selection set (n = 86) for τ = 0.41(balanced) and τ = 0.38 (high-sensitivity).

5-3 Results Comparison

For the final evaluation, the comparison is divided into two parts: an internal comparison justifying the ensemble approach over individual models, and an external comparison with SOTA literature.

5-3-1 Comparing Proposed Models (Base vs. Ensemble)

Table (5-1) illustrates the mean 5-fold performance of individual models against the ensemble framework.

Table 5. 1:Base Models vs. Ensemble Framework (5-Fold Mean)

Model	AUC	Accuracy	F1-Score	FP (Mean)	FN (Mean)
ResNet50-3D	0.5994	0.7649	0.8629	13.4	0.0
DualStreamMIL-3D	0.7990	0.8105	0.8822	9.0	1.8
SwinUNETR-3D	**0.9140**	0.8877	0.9227	2.8	3.6
Ensemble Framework	0.9114	**0.8807**	**0.9195**	**3.4**	**3.4**

Internal Analysis: Although the standalone SwinUNETR-3D model achieved an excellent AUC (0.9140), the ensemble framework proved **clinically superior** for two fundamental reasons:

1. **Error Balance:** The ensemble achieved a perfect balance (FN=3.4 vs. FP=3.4), whereas Swin skewed towards false negatives (3.6).
2. **Complementary Error Correction:** The ensemble successfully corrected **67%** (10 out of 15) of the cases misclassified by multiple base models.

MIL Ablation Study: We evaluated the efficacy of using Shannon Entropy to select

Figure 5. 5:Confusion matrices illustrating the classification performance of the ensemble model at different probability thresholds (τ = 0.41 and τ = 0.38).

Figure 5. 6:The main diagnostic dashboard of the proposed framework, illustrating the upload interface for the four required NIfTI modalities and the selection panel for the clinical operating threshold
Figure 5. 7:Confusion matrices illustrating the classification performance of the ensemble model at different probability thresholds (τ = 0.41 and τ = 0.38).

the top-16 slices. Entropy-based selection yielded an AUC of 0.7990 compared to 0.7310 for random selection, representing an **+8.0% relative improvement** and confirming the technique's success in filtering spatial noise.

5-3-2 Comparing Results with Previous Works

Based on the 25 studies extracted in the SLR (Chapter 3), Table (5-2) compares top literature models with our proposed framework.

Table 5. 2:Proposed Framework vs. Previous Works

Reference (Author, Year)	Architecture	AUC / Accuracy	Data Leakage Control (OOF)	Clinical Calibration (Platt/ECE)
Yin et al. [12]	CNN + Transformer 3D	AUC = 0.946	Patient-level Split (No OOF Meta-learner)	Eval only (Brier), no correction
Wen et al. [4]	3D U-Net	Acc = 94.4%	Absent (Potential Leakage)	Absent
Saluja & Trivedi [24]	Ensemble CNNs (2D)	AUC = 0.994	Absent(Potential Leakage)	Absent
Gutta et al. [1]	CNN + Transformer (2D)	F1 = 0.96	Patient-level (70/30 Split)	Absent
Proposed Framework	**D Ensemble 3 (CNN+Swin+MIL)**	**AUC = 0.912**	**Strict (Patient-level OOF)**	**Applied (Platt + Brier/ECE)**

Strengths and Weaknesses Analysis (Discussion):

- **Strengths of Previous Models:** Studies like Saluja (2025) and Yin (2025) achieved exceptionally high discriminative metrics (AUCs nearing 0.95–0.99).
- **Weaknesses of Previous Models:** As identified in our SLR, many of these models suffer from structural reliability flaws. Reaching high accuracy without strict patient-level separation (e.g., Saluja, Wen) or without utilizing Out-of-Fold (OOF) predictions in ensemble meta-learners (e.g., Yin) often leads to inflated performance and biased generalization. Furthermore, almost all cited studies relied on an uncalibrated, static threshold of 0.5.
- **Superiority of the Proposed Framework:** While our model may not hold the highest absolute mathematical AUC in the literature, it is arguably the most clinically safe and deployable system. Our project excels by providing:
    1. Unbiased evaluation through OOF stacking.
    2. Reliable, corrected probabilities via Platt Scaling.
    3. Dynamic clinical controllability, allowing oncologists to tune the threshold (0.38) to strictly minimize fatal HGG misclassifications (FN=3)—a feature entirely absent in previous research.

Future Improvements: Future work could enhance this framework by training on Multi-center datasets to boost absolute AUC values while strictly maintaining the robust reliability protocols established herein.

5.4 Clinical Deployment and User Interface

To fulfill the secondary research objective of providing a deployable clinical tool, the proposed ensemble framework was integrated into a comprehensive, interactive web-based diagnostic platform. This interface translates the complex theoretical pipelines—including modality fusion, meta-learning, and Platt scaling—into an accessible clinical decision support system.

Figure 5.5 illustrates the main diagnostic dashboard, which allows clinicians to upload the four required MRI modalities (T1, T1ce, T2, FLAIR) and select the desired clinical operating mode (Balanced vs. High-Sensitivity). Once the analysis is complete, the system presents the final predicted grade alongside the Platt-calibrated probability, as shown in **Figure 5.6**. To ensure clinical transparency and model explainability, the platform also provides a detailed breakdown of the ensemble's decision-making process (**Figure 5.7**), displaying the individual contributions of the SwinUNETR-3D, DualStreamMIL-3D, and ResNet50-3D base models, along with the applied logistic regression meta-learner formula.

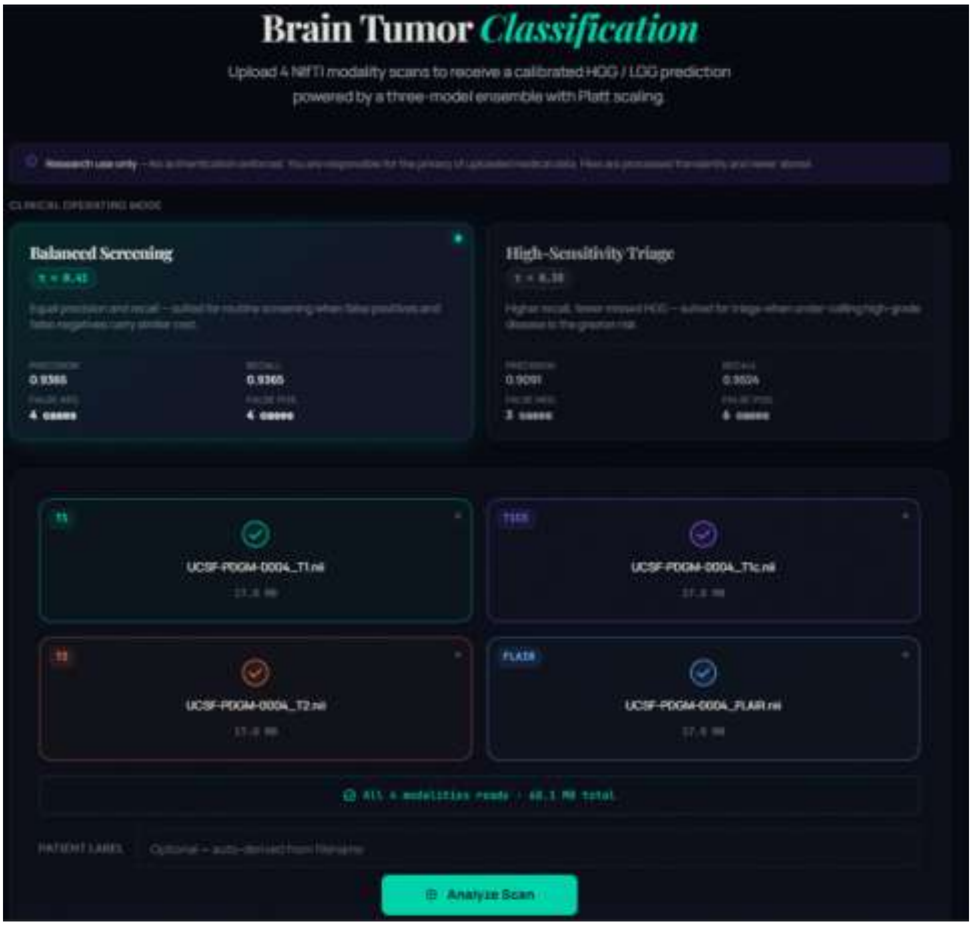

Figure 5. 8:The main diagnostic dashboard of the proposed framework, illustrating the upload interface for the four required NIfTI modalities and the selection panel for the clinical operating threshold (τ).

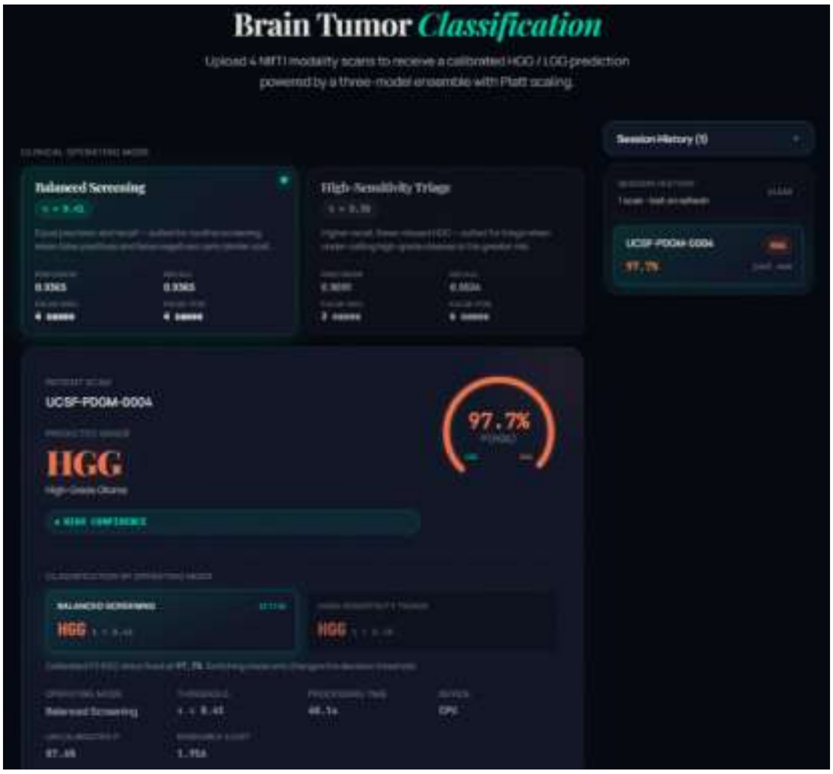

Figure 5. 10:The clinical prediction interface displaying the final diagnostic grade (HGG) and the Platt-calibrated confidence probability adjusted to the active threshold policy.

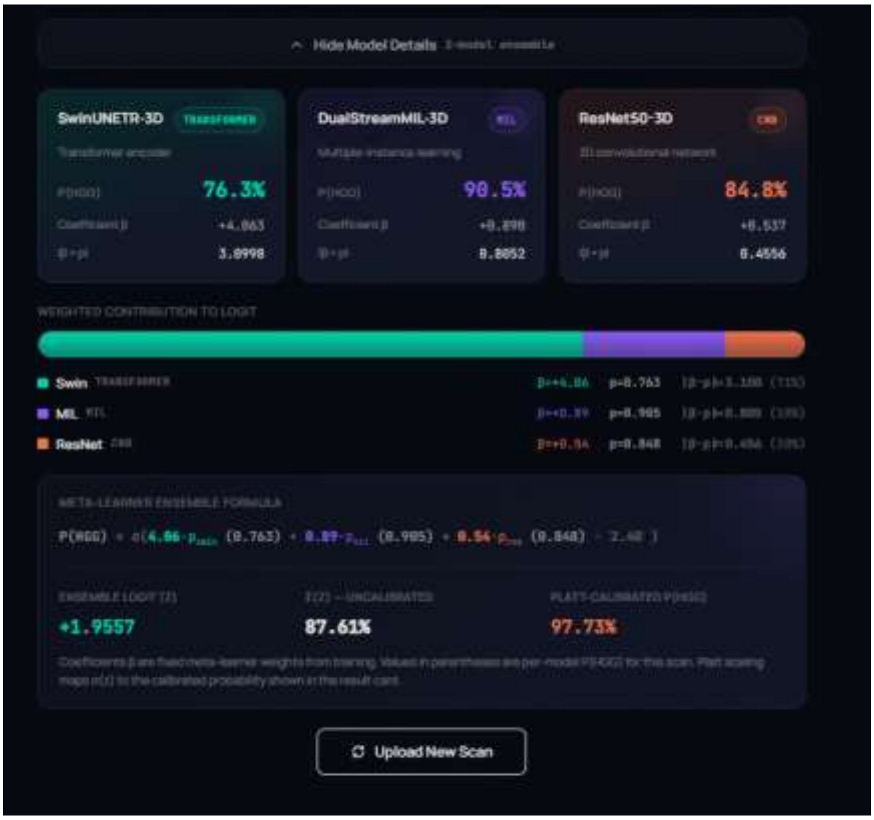

Figure 5. 9:The explainability and ensemble breakdown module, detailing the prediction probabilities of the three base architectures, their respective meta-learner coefficients (β), and the final calibration mapping.

Chapter 6: Conclusion

Introduction: This final chapter synthesizes the methodological and empirical milestones achieved throughout this research. The primary objective of this project was to design and construct a clinically reliable, calibration-aware, and controllable multimodal deep learning framework for brain tumor grade classification (HGG vs. LGG) utilizing multi-sequence 3D MRI. This chapter highlights the core conclusions drawn from our experimental validation, acknowledges the current challenges facing the evaluation pipeline, and outlines a comprehensive roadmap for future prospects aimed at transitioning this framework from an offline academic benchmark to a fully deployable clinical decision support tool.

6-1 Conclusions

Based on the extensive architectural deep-dive, strict leakage control audits, and quantitative results presented in Chapter 5, the following core conclusions are established:

1. **Synergy of Hybrid Inductive Biases:** The integration of volumetric deep learning (ResNet50-3D), hierarchical self-attention (SwinUNETR-3D), and slice-level multiple instance learning (DualStreamMIL-3D) proves that combining distinct architectural paradigms yields superior robust representations. The ensemble meta-learner effectively exploited these complementary signals, successfully correcting **67%** of cases where multiple individual base models failed.
2. **Methodological Rigor via Data Leakage Control:** Implementing a strict patient-level splitting protocol (5-Fold Stratified Patient-Level CV) paired with training the meta-learner exclusively on Out-of-Fold (OOF) predictions is verified as an absolute requirement for unbiased evaluation. This rigor protects the framework from the optimistic performance inflation prevalent in literature where slice-level leakage goes unchecked.
3. **Clinical Priority of Probability Calibration:** High discriminative performance (AUC) is insufficient for medical safety without probabilistic reliability. The application of post-hoc Platt scaling on the ensemble's log-odds successfully corrected model overconfidence, yielding a **26.9% reduction in Expected Calibration Error (ECE)** and a **16.8% reduction in Brier Score**, thereby aligning the framework’s numeric confidence with true empirical clinical frequencies.
4. **Flexibility of Clinically Tunable Thresholds:** Moving away from the static, unverified 0.5 decision threshold is vital for clinical risk management. By executing a threshold sweep on independent out-of-fold data, the framework established two operational modes: a balanced operating point (**0.41**) optimizing holistic screening accuracy, and a high-sensitivity operating point (**0.38**) that strictly minimized fatal False Negatives for HGG cases to just **3 patients**, achieving a clinically crucial Recall of **0.9524**.
5. **Efficacy of Information-Theoretic Feature Selection:** Information-driven slice filtering via Shannon Entropy for the MIL pipeline provided an empirical **+8.0% relative improvement in AUC** compared to random slice sampling, mathematically validating that eliminating uninformative background

volumetric noise prior to deep aggregation significantly stabilizes representation learning.

6-2 Challenges and Future Prospects

While the framework achieves robust design verification, several structural and technical challenges remain, highlighting promising pathways for future research:

I. Current Challenges

1. **Single-Cohort Development Constraint:** The current model was trained and tuned exclusively on the BraTS 2018 cohort (285 patients). Although strictly validated, this localized optimization introduces vulnerability regarding inter-institutional generalizability when exposed to diverse clinical environments with varying scanner hardware and regional acquisition protocols.
2. **Computational Overhead of Volumetric 3D Context:** Processing full 3D spatial volumes (128^3) across quadratic self-attention layers in SwinUNETR-3D and dense volumetric convolutions requires extensive GPU VRAM and a high computational budget. This overhead presents an operational bottleneck for real-time inference or deployment in standard hospital systems with limited hardware resources.
3. **Isolation from Non-Imaging Modalities:** The proposed framework operates purely in a radiological vacuum, analyzing four MRI channels (T1, T1ce, T2, FLAIR). In actual neuro-oncological practice, clinical decisions are a mosaic that combines imaging with qualitative patient history, symptomatology, and genomic pathways—modalities that remain unintegrated in the current architecture.

II. Future Prospects

1. **Multi-Center External Validation:** The highest priority for subsequent research is conducting extensive validation on completely independent, external clinical datasets harvested from local hospital networks. This will rigorously test the generalizability of our calibrated probabilities and the robustness of the established operational thresholds (0.41 / 0.38) under true clinical heterogeneity.
2. **Multimodal Clinical-Genomic Stacking:** Expanding the architecture to process a holistic clinical bag by integrating raw imaging with patient text metadata and genomic biomarkers (e.g., IDH mutation status and 1p/19q co-deletion). Utilizing biomedical Large Language Models (LLMs) like clinical BERT to encode textual data will enable a comprehensive multi-task pipeline linking segmentation, grading, and survival prediction.
3. **Transition to Highly Efficient Linear State-Space Models:** Future work will explore substituting heavy transformer attention layers with emerging linear-time architectures, such as 3D State-Space Models (3D Mamba). This paradigm shift can circumvent the quadratic complexity of standard self-attention while preserving long-range volumetric context, drastically reducing VRAM footprints and accelerating clinical inference.

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Appendices

Appendix A: Submitted Research Manuscripts

Reliable Glioma Grading via Leakage-Controlled Multimodal 3D Ensemble Learning

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Abstract. High-grade gliomas (HGG) require accurate stratification from low-grade gliomas (LGG) using multi-modal MRI. While deep learning models achieve strong discrimination performance, insufficient leakage control and poorly calibrated probabilities can limit clinical reliability. We present a leakage-controlled multimodal 3D ensemble learning framework for reliable glioma grading. The system integrates ResNet50-3D, SwinUNETR-3D, and entropy-based DualStreamMIL-3D within a strict 5-fold patient-level cross-validation protocol using out-of-fold stacking. A logistic regression meta-learner is trained exclusively on out-of-fold predictions with balanced class weights. Post-hoc Platt scaling is applied on a dedicated calibration split, followed by held-out threshold selection to enable clinically controllable operating points. On BraTS 2018 ($n = 285$), the ensemble achieved a mean AUC of 0.9114 ± 0.0423 across folds and a full out-of-fold AUC of 0.9126, compared to 0.9065 for the strongest single backbone. Calibration improved probability reliability (ECE = 0.116 on full OOF) while preserving discrimination. At the balanced threshold (0.41, held-out $n = 86$), the model achieved precision and recall of 0.9365 with controlled false negatives; a high-sensitivity setting (0.38) further reduced false negatives. These results demonstrate that strict leakage control combined with post-hoc calibration enables clinically reliable ensemble deployment without inflating performance estimates.

Keywords: Glioma grading · Multimodal MRI · 3D Deep Learning · Ensemble Learning · Calibration · Data Leakage Control

1 Introduction

Accurate glioma grading is clinically critical because missed HGG cases can delay urgent treatment, while LGG often follow different management pathways. Although multi-modal MRI (T1, T1ce, T2, FLAIR) enables strong discrimination, deployment requires more than AUC: predicted probabilities must be reliable and operating thresholds must be selected without leakage. In stacking and post-processing pipelines, reusing validation predictions for meta-learning, calibration, or threshold tuning can introduce optimistic bias [11, 12]. We therefore frame glioma grading as a deployment-oriented task requiring (i) leakage-controlled prediction aggregation, (ii) calibrated confidence estimates, and (iii) explicitly controllable operating points for FN-sensitive use.

Figure A. 1:Title page of the manuscript submitted to the MICCAI 2026 Mentorship Program

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From Accuracy to Clinical Reliability: A Systematic Review of Deep Learning in Glioma Grading

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ABSTRACT In the domain of neuro-oncology, the precise classification of gliomas into high-grade and low-grade categories is essential for formulating effective treatment protocols and determining patient prognosis. Although deep learning methodologies applied to Magnetic Resonance Imaging (MRI) have shown remarkable discriminative precision, their integration into dependable clinical instruments is frequently impeded by methodological shortcomings, including uncalibrated predictive probabilities and latent data leakage. This manuscript presents a comprehensive systematic literature review of research focused on deep learning-based glioma grading published between 2018 and 2026.
Adhering to rigorous systematic review protocols, we investigated 25 primary studies systematically filtered from an initial collection of 1,676 publications. This review offers a structured synthesis of the research landscape by (i) evaluating multimodal MRI fusion and image pre-processing pipelines, (ii) contrasting computational architectures ranging from conventional 2D/3D Convolutional Neural Networks (CNNs) to Vision Transformers and ensemble models, (iii) scrutinizing structural integrity concerning patient-level data leakage, and (iv) appraising model calibration alongside clinical risk mitigation strategies.
Our analysis highlights a distinct methodological shift toward spatially comprehensive, multimodal, and hybrid 3D networks. Nevertheless, the implementation of these modalities is frequently suboptimal, as numerous studies still depend on 2D slice-based evaluations that omit vital volumetric context. Moreover, despite reporting exceptional accuracy metrics, the clinical applicability of these algorithms is severely restricted by profound vulnerabilities: erratic patient-level data splitting practices, a near-total deficiency in post-hoc probability calibration (e.g., Platt Scaling), and the utilization of static decision thresholds that disregard the perilous risks associated with False Negatives in high-grade glioma detection.
To navigate these impediments, this review pinpoints critical research voids and proposes future trajectories for establishing clinically robust systems. We particularly emphasize the necessity of strict Out-of-Fold (OOF) stacking, entropy-driven Multiple Instance Learning (MIL), and adaptable operating thresholds. Ultimately, this study synthesizes recent progress in automated glioma grading into a cohesive framework, offering strategic insights for engineering more resilient, accurately calibrated, and clinically practical artificial intelligence diagnostic applications.

INDEX TERMS Glioma grading, Systematic literature review, Deep learning, Multimodal MRI, Data leakage, Model calibration, Ensemble learning, Clinical reliability.

I. INTRODUCTION

IN the field of neuro-oncology, brain tumors—particularly gliomas—represent some of the most formidable malignancies. The precise staging of these neoplasms, differentiating High-Grade Gliomas (HGG) from Low-Grade Gliomas (LGG), constitutes a pivotal phase in establishing therapeutic regimens and estimating patient outcomes. Although Magnetic Resonance Imaging (MRI) serves as the primary non-invasive diagnostic modality, traditional manual assessments are inherently labor-intensive, highly contingent upon the clinician's proficiency, and vulnerable to subjective variance.

In contrast to conventional diagnostic protocols, deep learning algorithms have fundamentally transformed the automated evaluation of medical imaging. Because individual MRI sequences (such as T1, T1ce, T2, and FLAIR) reveal unique histological properties, the fusion of multimodal data

Figure A. 2:Title page of the manuscript submitted to IEEE

Appendix B: Detailed Systematic Literature Review (SLR) Table

This appendix contains the comprehensive data extraction matrix for the 25 studies accepted through the SLR process described in Chapter 3.

Figure B. 1:Systematic Literature Review Summary Table

ID	Title	Authors	Year	Input (MRI Sequences)	Input Category (Single-modality or Multi-modality)	Type of Input (2D/3D)	Output (HGG/LGG)	Pre-processing	Pre-processing Category (Basic or Advanced)	Representation Approach	Solving Approach	Methodology Category (Standard DL or Transformer or Ensemble or Hybrid.)	Pipeline Description	Leakage Control	Calibration	Reliability Category (Low Reliability or High Reliability)	Results	Main Limitation / Notes
1	Improving glioma grade classification with a hybrid CNN-transformer model	Sreedevi Gutta, Shyam Sundhar Yathirajam	2026	BraTS 2020 Dataset. MRI modalities: T1CE, T2, FLAIR (T1 was excluded)	Multi-modality	2D (Tumor-relevant 2D slices extracted from 3D MRI volumes)	Glioma Grade Classification (HGG vs. LGG)	Co-registration, skull-stripping, resampling to 1 mm³, slice selection (discarding non-tumor slices), Z-score normalization, and pixel intensity clipping to [-5, 5].	Advanced	3-channel tensor (T1CE, T2, and FLAIR stacked as RGB channels)	Hybrid architecture combining a lightweight Convolutional Neural Network (CNN) with a Vision Transformer (DeiT-Tiny) backbone.	Hybrid	1. Select tumor-containing 2D slices from the MRI dataset. 2. Stack T1CE, T2, and FLAIR as RGB channels, normalize, and clip intensities. 3. Pass the input through a CNN branch to extract local spatial features. 4. Concurrently pass the input patches through a DeiT-Tiny transformer branch to capture global contextual patterns. 5. Concatenate the CNN and Transformer embeddings into a unified vector. 6. Perform binary classification (LGG vs. HGG) using dense layers. 7. Aggregate individual slice predictions for each patient via majority voting.	Stratified Patient-level split. Data was split 70% train and 30% test at the patient level ensuring slices from the same individual never appeared in both subsets. Patient-level predictions were used via majority voting.	Not Mentioned.	High Reliability	Test Set Results for Hybrid Model: F1-Score: 0.96, Accuracy: 0.97, Precision: 0.95, Recall: 0.97.	Limitations: Evaluated solely on BraTS 2020 (needs external validation), and assumes all 3 MRI modalities are perfectly available without addressing missing modalities. Innovation: Highly efficient clinically (13 ms inference per slice, 89 MB model size) and features strong interpretability combining Grad-CAM (for CNN edges) and Attention Maps (for global context).
2	A Novel Hybrid Model for Brain Tumor Analysis Using Dual Attention AtroDense U-Net and Auction Optimized LSTM Network .	S. K. Rajeev, M. Pallikonda Rajasekaran, R. Kottaimalai, T. Arunprasath, Nisha A. V., Abdul Khader Jilani Saudagar .	2026	BraTS 2019 and BraTS 2020 datasets. MRI modalities: T1, T1ce, T2, and T2-FLAIR	Multi-modality	2D (The segmentation utilizes a 2D Atrous encoder block).	Tumor Segmentation, followed by Glioma Grading Classification (HGG vs. LGG)	Gaussian bilateral filter for noise reduction and edge preservation. (Dataset also inherently includes skull-stripping and co-registration).	Advanced	4-channel tensor for segmentation input; 1D sequential radiomic feature vector for LSTM classification.	Hybrid architecture consisting of DA-AtroDense U-Net for segmentation, combined with handcrafted feature extractors (KED, GLCM), CMO optimizer, and Auction-Optimized LSTM (AOHLN) for classification.	Hybrid	1. Pre-process MRI scans using a Gaussian bilateral filter. 2. Segment the tumor regions using the proposed DA-AtroDense U-Net. 3. Extract texture and structural radiomic features from the segmented areas using Kirsch Edge Detector (KED) and GLCM. 4. Select the most relevant features to reduce redundancy using Cat-and-Mouse Optimization (CMO). 5. Classify the tumor (HGG vs LGG) using an Auction-Optimized Hybrid LSTM Network (AOHLN).	Random split. The authors explicitly stated: "Patient-level separation was not explicitly enforced in the original implementation"	Not Mentioned	Low Reliability.	Segmentation: DSC = 0.9907. Classification (BraTS 2020): Accuracy = 99.04%, F1-Score = 99.30%. Classification (BraTS 2019): Accuracy = 98.81%, F1-Score = 99.22%.	Limitations: Did not strictly enforce patient-level data splitting (high risk of data leakage). Uses handcrafted feature extraction instead of end-to-end deep learning classification, leading to high computational time. Innovation: Novel integration of Cat-and-Mouse Optimization for feature selection and Auction-Based Optimization for LSTM weights.

3	A Comparative Study of IVIM-MRI Fitting Techniques in Glioma Grading: Conventional, Bayesian, and Voxel-Wise and Spatially-Aware Deep Learning Approaches.	Misha P. T. Kaandorp, Andras Jakab, Christian Federau, Peter T. While.	2026	Synthetic fractal-noise data (for training) + Preoperative DWI from 20 glioma patients (for testing). Modality: Diffusion-Weighted MRI (DWI) with 16 b-values.	Single-modality (Only DWI used)	2D (Patch-based training and inference on slices))	Glioma Grading Classification (Grade 4 vs. Grade 2/3), matching HGG vs. LGG	Generation of synthetic training data. For in-vivo data: extracting IVIM parameters (Diffusion D, pseudo-diffusion D*, and perfusion fraction f) from 16 b-value DWI signals.	Advanced	Parametric maps (D, D*, f) generated from 16-channel b-value DWI sequences	Spatially-aware Transformers (NATTEN-17, SA-17) compared against Conventional (LSQ, SEG), Bayesian (BSP, FBM), and Voxel-wise DL (IVIM-NET).	Transformer	1. Train Transformer models purely on synthetic fractal-noise data (no real patients). 2. Input 16 b-value DWI scans of real patients into the trained models. 3. Generate and extract IVIM parameter maps (D, D*, f). 4. Extract parameter values from the whole-tumor ROI. 5. Classify tumors into Grade 4 or Grade 2/3 and evaluate performance using ROC-AUC.	Strict Zero Data Leakage. The DL models were trained entirely on synthetic fractal-noise data. The 20 in-vivo patient datasets were exclusively used for testing.	Not Mentioned	High Reliability	The SA-17 Transformer achieved the highest ROC-AUC for D* = 0.78 (0.80 with Gaussian prior), D = 0.71, and f = 0.65. It significantly outperformed conventional LSQ (AUC D*=0.62).	Limitations: Very small in-vivo cohort (20 patients), single-center study, class imbalance (12 Grade-4 vs 8 Grade-2/3), and the model relies on a fixed 16 b-value MRI protocol (requires retraining if the protocol changes). Innovation: Successfully training Transformers solely on synthetic data, eliminating the need for large annotated clinical datasets, yet generalizing well to in-vivo scans.
4	A deep ensemble learning framework for glioma segmentation and grading prediction	Liang Wen, Hui Sun, Guobiao Liang & Yue Yu	2025	BraTS 2020 Dataset. Modalities: T1, T1ce, T2, FLAIR	Multi-modality	3D (Volumes cropped to 128x128x128)	Glioma Segmentation (Tumor boundaries) and Grading Classification (HGG vs. LGG)	Cropping images to 128x128x128, Bias field correction using SimpleITK, Z-score Standardization (normalization), and Data Augmentation (random flipping, rotation, translation).	Advanced	4-channel 3D tensor (R^(H × W × D × C), where C=4 modalities)	Multi-task Deep Ensemble Learning based on 3D U-Net. It uses a shared encoder with 3D Asymmetric Convolution Blocks (ACB) and Dual-Domain Attention (DA), combined with a dual-branch decoder.	Hybrid	1. Preprocess 4-modality 3D MRI data (bias correction, crop, normalize). 2. Pass the data through a shared 3D U-Net encoder utilizing 3D Asymmetric Convolution Blocks (ACB) and Dual-Domain Attention (CA+SA) to extract global/local features. 3. Feed the shared features into a dual-branch decoder. 4. Branch 1 uses upsampling and skip connections to generate voxel-level tumor segmentation. 5. Branch 2 applies Global Average Pooling (GAP) and Global Maximum Pooling (GMP), concatenates them, and uses a linear classifier layer for tumor grading (HGG vs LGG). 6. Optimize end-to-end jointly using an adaptive weighted composite loss function ($\alpha=0.5$).	Not Mentioned (The paper mentions a 2:1 train-to-test split but does not explicitly declare a strict patient-level separation to prevent leakage).	Not Mentioned	Low Reliability	Grading Classification: Accuracy = 94.42%, Recall = 91.81%, Specificity = 97.49%. Segmentation: Average Dice Score = 87.66% (WT = 91.28%, TC = 85.82%, ET = 85.88%).	Limitations: High computational complexity limiting real-time usage; relies entirely on the BraTS 2020 dataset (limiting generalization); performance is highly sensitive to missing or corrupted MRI modalities. Innovation: Proposes the novel "GSG-UNet" combining multi-task learning, Asymmetric Convolution Blocks (ACB) to capture complex tumor shapes, Dual-domain Attention, and a dynamic composite loss function.
5	RGNN3D: A hybrid radiomic graph neural network for 3D MRI glioma grading	Md Aiyub Ali et al.	2026	BraTS 2019 and BraTS 2020 datasets. Modalities: T1, T2, T1CE, FLAIR	Multi-modality (Evaluated independently for each modality)	3D (3D MRI scans processed into 112-D radiomic feature vectors)	Glioma grading classification (GBM [representing HGG] vs. LGG)	Identification of Regions of Interest (ROI), and extraction of 112 radiomic features using PyRadiomics (covering First Order, Shape, GLCM, GLRLM, GLSZM, NGTDM, GLDM). BraTS inherent preprocessing (skull-stripping, co-registration) is used.	Advanced	Graph representation. Node feature matrix X ∈ ℝ^(704 × 112) (112-D vector per patient). Identity Adjacency Matrix A = I_704.	Hybrid architecture combining Graph Convolutional Networks (GCN) with Long Short-Term Memory (LSTM) cells.	Hybrid	1. Acquire 3D MRI scans and identify tumor ROIs. 2. Extract 112 radiomic features using PyRadiomics. 3. Construct a graph where each patient is an independent node. 4. Feed features into a Feed-Forward Network (FFN). 5. Pass through GCN layers integrated with LSTM cells to update node embeddings. 6. Use a dense layer with Softmax for final classification. 7. Apply LIME for model explainability.	Strict Patient-level separation. The graph strictly uses an Identity Adjacency Matrix (A = I_704) ensuring self-loops only. This completely isolates patients during message passing, preventing any cross-patient information leakage. 10-fold cross-validation is used.	Not Mentioned	High Reliability	98.58% (best on T1 and T2), F1-Score: 99.10%, Precision: 100%, Recall: 98.23%.	Limitations: Evaluated on a limited dataset (BraTS 19-20); performance drops on FLAIR modality; needs validation in real clinical environments. Innovation: Novel integration of LSTM within GCN to process radiomics without data leakage. Excellent use of LIME (Explainable AI) to identify the top 3 radiomic biomarkers driving the grading decision.

6	MoRE-Net: An Interpretable and Modality-robust Model for Brain Tumor Grading	Binghua Li et al.	2026	BraTS 2020 dataset and ReMIND dataset. Modalities: T1, T1CE, T2, FLAIR	Multi-modality	3D (Volumes resized to 128 × 128 × 96)	Glioma grading (HGG vs. LGG)	Isotropic resampling (1 mm³), skull-stripping, cropping/padding to 192 × 192 × 144, resizing to 128 × 128 × 96.	Advanced.	Independent per-modality feature embeddings (Mamba-based) fused into a combined embedding.	Hybrid: Interpretable Prototypical Part Network (MProtoNet variant) + Mamba-based encoders + Multimodal teacher with alignment loss.	Hybrid (Integrates Transformer-like components, CNN, and Interpretability blocks)	1. Preprocess 3D MRI volumes. 2. Encode each modality independently using Mamba-based encoders. 3. Apply random modality dropout during training for robustness. 4. Use an online multimodal teacher (CNN-based early fusion) to guide per-modality encoders via alignment loss. 5. Process fused features through a Prototypical Part Network module. 6. Generate final classification and attribution maps for interpretability.	5-fold cross-validation with subject-stratified splitting	Not explicitly mentioned (Brier score is used to assess calibration)	High Reliability	BraTS2020: Average BAC: 73.5%, Activation Precision (AP): 61.2%	Limitation: Trained under fully observed multimodal assumption; struggles with very severe modality loss. Innovation: First to simultaneously address robustness (missing modalities) and interpretability in 3D glioma grading; introduces Mamba-based encoders for global context modeling.
7	Enhancing Brain Tumor Classification and Generalization Using DDPM-Generated MRI, Mutual Information and Ensemble Learning	Yael H. Moshe et al.	2026	BraTS 2019 dataset + TASMC local dataset. Modalities: T1WI, T1WI+C, T2WI, FLAIR	Multi-modality	2D (Central slices extracted from 3D volumes)	Glioma grading (HGG vs. LGG)	Cropping around the tumor center, resizing to 96x96, Z-score normalization	Basic	4-channel tensor (accommodating the four MRI sequences)	DDPM (Denoising Diffusion Probabilistic Models) with Mutual Information (MI) regularization for synthetic image generation + 2D ResNet-152 for classification + Ensemble Learning (5-fold cross-validation).	Ensemble	1. Select central tumor slices from 3D MRI. 2. Train DDPM (with and without MI) to generate synthetic images. 3. Train 2D ResNet-152 on four setups: Real, Real+Augmentation, Real+DDPM, Real+DDPM+MI. 4. Use 5-fold ensemble to average model outputs for final grade prediction.	Subject-level splitting (80/20 train/test split)	Not mentioned (Brier score used for evaluation only)	High Reliability	Accuracy (BraTS-to-TAMSC): 0.89, (TAMSC-to-BraTS): 0.85	Limitation: Focused on 2D slices (ignoring volumetric context); relies on limited patch size (96x96); high computational demand. Innovation: Introduces MI regularization in DDPM to increase image diversity and reduce overfitting (memorization), which significantly improves generalization across different datasets.
8	Enhancing Glioma Classification in Magnetic Resonance Imaging Using Vision Transformers and Convolutional Neural Networks	Marco Antonio Gómez-Guzmán et al.	2026	BraTS 2019 "best-slice" dataset. Modalities: T1, T1CE, T2, FLAIR	Multi-modality	2D (Single "best-slice" images extracted from 3D volumes).	Glioma grading (HGG vs. LGG)	Image resizing (224x224), color adjustment, rotation, and random horizontal/vertical flipping (dynamic augmentation).	Basic	3-channel RGB images (converting single MRI slice to RGB by repeating channels)	Evaluation of six architectures: DeiT3_base, Inception_v4, Xception41, ConvNextV2_tiny, Swin_tiny, and EfficientNet_B0.	Transformer/Standard DL (Comparison study of multiple architectures)	1. Extract best-slice images from BraTS 2019. 2. Preprocess and apply random dynamic augmentation. 3. Train six pre-trained DL models using Transfer Learning. 4. Evaluate using 3-fold cross-validation. 5. Perform final classification and interpret results via Grad-CAM.	Patient-level splitting (Training and testing folders contain images from different patients)	Not Mentioned	Low Reliability	DeiT3: 99.40% Accuracy; Inception_v4: 99.21% Accuracy	Limitation: Focus on 2D slices (excludes volumetric context); relies on best-slice selection which may miss multi-slice features; lack of external clinical validation. Innovation: Comprehensive benchmarking of 6 modern DL architectures and detailed computational cost analysis (GPU usage, inference time).

9	Context aware deep learning for brain tumor segmentation, subtype classification, and overall survival prediction using radiology images	Linmin Pei et al.	2020	BraTS 2019 dataset + CPM-RadPath 2019 dataset. Modalities: T1, T1ce, T2, FLAIR	Multi-modality	3D	Tumor segmentation, Glioma subtype classification, and Overall survival prediction	Co-registration, skull-stripping, noise reduction, and Z-score normalization for brain regions.	Advanced	4-channel tensor (4 × 160 × 192 × 128)	Hybrid architecture: 3D CANet (Context-Aware Network) for segmentation, 3D CNN for subtype classification, and Hybrid ML/DL for survival prediction.	Hybrid	1. Preprocess mMRI scans. 2. Apply CANet for tumor segmentation. 3. Feed tumor segments into a 3D CNN for classification. 4. Extract features from CANet frontend, apply LASSO for selection, and use linear regression for survival prediction.	Patient-level splitting (Standard challenge-provided splits)	Not Mentioned	High Reliability	Segmentation DSC: 0.821–0.895 (depending on region); Classification Accuracy: Ranked 2nd in CPM-RadPath challenge.	Limitation: Data imbalance and relatively small sample size in some tasks. Innovation: Integrated framework for multiple correlated tasks (segmentation, classification, survival) rather than isolated analysis.
10	Classification of brain tumours in MR images using deep spatiospatial models	Soumick Chatterjee et al.	2022	BraTS 2019 dataset (for tumors) and IXI dataset (for healthy brains). Modality: T1ce (T1-contrast-enhanced).	Single-modality (T1ce).	3D (Volumetric MRI scans).	Glioma grading (HGG vs. LGG) + Healthy brain classification.	Brain extraction (BET2), intensity normalization (99.5 percentile), and resampling to 2mm isotropic resolution.	Advanced.	3D volumetric MRI tensors.	Hybrid Spatiospatial architectures (ResNet (2+1)D, ResNet Mixed Convolution) + Transfer Learning.	Hybrid (treating slice dimension as pseudo-temporal dimension).	1. Preprocess MRI scans. 2. Resample to 2mm. 3. Initialize models with Kinetics-400 pre-trained weights. 4. Train using spatiospatial architectures. 5. Perform 3-fold cross-validation. 6. Classify into LGG, HGG, or Healthy.	Patient-level splitting (using 3-fold cross-validation).	Not Mentioned.	High Reliability.	Best model (Pre-trained ResNet Mixed Conv): Macro F1-score: 0.9345, Test Accuracy: 96.98%.	Limitation: Focus on T1ce only (might miss multi-sequence benefits); limited maintenance of 3D spatial relationship (pseudo-temporal treatement). Innovation: Novel application of "spatiospatial" models (usually for action recognition in video) to 3D MRI volumes to reduce parameters and training costs while maintaining high accuracy.
11	Biologically interpretable multi-task deep learning pipeline predicts molecular alterations, grade, and prognosis in glioma patients	Xuewei Wu et al.	2024	Private datasets (FAHZU, HPPH) and Public datasets (TCGA, UCSF, EGD). Modalities: MRI (T1, T1ce, T2, FLAIR).	Multi-modality.	3D (Volumes resized/padded to 88 × 112 × 88).	IDH mutation status, 1p/19q co-deletion status, Histological grade (HGG/LGG components), and Overall Survival.	Co-registration, brain extraction (BET), intensity standardization, and Z-score normalization.	Advanced.	512 shared deep feature vectors extracted from a 3D ResNet-10 backbone.	Multi-task Deep Learning (MDL) using a shared 3D ResNet-10 encoder with multiple task-specific heads.	Hybrid (Multi-task deep learning architecture).	1. Preprocess mMRI data. 2. Automated tumor segmentation via pre-trained mmFormer. 3. Extract shared features using 3D ResNet-10. 4. Concurrent prediction for molecular markers, grade, and survival via dedicated branches.	Patient-level split across multiple independent institutions.	Yes (Calibration curves were plotted for survival prediction).	High Reliability.	AUC for IDH: 0.892–0.903, 1p/19q: 0.710–0.894, Grade: 0.850–0.879; OS Prediction (C-index): 0.671–0.723.	Limitation: Retrospective design; multi-omics analysis limited to TCGA. Innovation: First multi-task MDL concurrently predicting markers, grade, and prognosis with biological interpretability via multi-omics.

12	Integrating deep learning and radiomics for preoperative glioma grading using multi-center MRI data	Shi Yin et al.	2025	Multi-center data (5 tertiary centers). Modalities: T1, T1-contrast, T2, FLAIR	Multi-modality	3D (Volumetric MRI scans)	Glioma grading (HGG vs. LGG)	Brain extraction (HD-BET), registration, intensity normalization (WhiteStripe/Z-score), histogram matching.	Advanced	Dual-stream: 1,423 quantitative radiomic features + 512 deep learning feature vector	3D CNN backbone + Transformer-based attention module + Radiomics feature selection (RFECV) + Attention-based feature fusion.	Hybrid	1. Preprocess multimodal MRI. 2. Automated segmentation into 5 ROIs (ET, NET, ED, PPZ, DPZ) using 3D U-Net. 3. Extract radiomic features (1,423) and deep features (CNN + Transformer). 4. Habitat analysis to quantify spatial heterogeneity. 5. Integrate all streams via an attention-based fusion module. 6. Final classification via classifier head.	Patient-level split (multicenter study with independent external cohort)	Yes (Calibration curves were used and Brier scores were reported)	High Reliability	Integrated model: AUC = 0.946 (Internal), 0.921 (External Validation)	Limitation: Retrospective design, potential selection bias; computationally demanding dual-stream architecture. Innovation: Habitat analysis (spatially detailed heterogeneity) + dual-stream fusion (Radiomics + DL) + Domain adaptation for scanner variations.
13	Efficient hybrid fuzzy weighted 3D FCNN with TSO PSO optimization for accurate multi modal MRI brain tumor classification	Liu Feng, Guo Hao & Wang Jian	2025	BraTS 2019, BraTS 2020, and a subset of BraTS 2021. Modalities: T1, T2, FLAIR, and T1ce.	Multi-modality.	3D (Volumetric MRI scans).	Glioma grading (HGG vs. LGG).	Skull-stripping, normalization, resampling to 128 × 128 × N, and histogram-based contrast enhancement.	Advanced.	128-dimensional feature vectors extracted from 3D-FCNN.	Hybrid architecture: 3D-FCNN with Interval Type-2 Fuzzy Weighting + Hybrid TSO-PSO Optimization + Classification via SVM or SoftMax.	Hybrid.	1. Preprocess multimodal 3D MRI. 2. Initialize filter weights using T-S Fuzzy logic. 3. Extract features via 3D-FCNN. 4. Optimize network weights/parameters using TSO-PSO algorithm. 5. Classify using SoftMax or SVM.	Patient-level split (Stratified training, validation, and testing sets).	Not Mentioned.	High Reliability.	Accuracy: 98.1%, Sensitivity: 98.9%, Specificity: 95.0%, Dice Score: 0.987.	Limitation: Evaluation restricted to BraTS datasets; current model lacks real-time inference testing. Innovation: Fuzzy weight generation (reducing trainable parameters by ~92.6% and training time by 6x); novel hybrid metaheuristic optimization (TSO-PSO).
14	DGIResNet-based multi-gene classification method for adult diffuse glioma using MRI	Hongwei Zhang et al.	2025	TCIA (The Cancer Imaging Archive) and The First Affiliated Hospital of Chongqing Medical University. Modalities: T1C (T1-contrast) and T2F (T2-FLAIR).	Multi-modality (Concatenation of T1C and T2F).	2D (Slices extracted from 3D volumes).	5 categories of adult diffuse glioma (including IDH mutation, 1p/19q codeletion, and WHO grading).	Normalization, Resizing to 224x224, Data augmentation (Flip, Rotation).	Basic.	Feature-map based representation using grouped and dilated convolutions.	Hybrid CNN (Custom DGIResNet: Inverted residual + Grouped convolution + Dilated convolution).	Standard DL (Customized CNN architecture).	1. Preprocess T1C and T2F slices. 2. Fuse images via channel concatenation. 3. Feature extraction via DGIResNet module. 4. Multi-gene classification using the extracted feature maps.	Not Mentioned (The paper mentions 7:2:1 splitting but does not explicitly detail patient-level separation to prevent leakage).	Not Mentioned.	Low Reliability.	Accuracy: 0.7332 (Validation set), AUC: 0.9279, F1: 0.7150	Limitation: Smaller sample size for some genetic subtypes; reliance on 2D slices loses 3D spatial context. Innovation: DGIResNet architecture optimizes channel/spatial scales with grouped/dilated convolutions for efficient multi-gene classification.

15	GMRNet: Deep Residual Network–Based Radiopathomic Glioma Classification with the Fusion of Multi-omics Data	Hongwei Zeng et al.	2025	HMG (Private) and OAG (Public) datasets. Modalities: T1C (T1-contrast) and T2F (T2-FLAIR) MRI + WSI (Whole Slide Imaging).	Multi-modality (Radiopathomic fusion).	2D (Patches extracted from MRI slices and WSI).	5 glioma subtypes (OII, AII, AA, AO, GBM).	Affine transformations, contrast enhancement, intensity normalization.	Advanced.	Multi-scale feature concatenation (Radiopathomic fusion of MRI and WSI deep features).	Deep Residual Network (ResNet-50 based) with dual-stream architecture.	Hybrid (Fusion of Radiomic and Pathomic deep learning branches).	1. Data augmentation. 2. Dual-stream feature extraction using ResNet-50 backbones (one for MRI, one for WSI). 3. Feature-level fusion via concatenation. 4. Bottleneck layer for dimensionality reduction. 5. Multilayer classification network.	Patient-level leave-one-out cross-validation (LOOCV).	Not Mentioned.	High Reliability.	Accuracy: 93.65% (HMG dataset), 86.47% (OAG dataset). F1-score: 91.36% (HMG).	Limitation: AO and OII subtypes confusion due to feature space overlap. Innovation: Radiopathomic fusion (Integrating histology features with MRI deep features) + Margin-based top-K pooling for robust patient-level aggregation.
16	An Attention-Guided CNN Framework for Segmentation and Grading of Glioma Using 3D MRI Scans	Prasun Chandra Tripathi and Soumen Bag	2023	BraTS 2019 dataset (Segmentation) and TCIA datasets (Classification - TCGA-GBM, TCGA-LGG, TCGA-1p19qdeletion). Modalities: T1, T2, T1C, FLAIR.	Multi-modality (Multi-channel MRI input).	3D (3D MRI volumes).	Glioma grading (HGG vs. LGG), 1p/19q status, and IDH mutation status.	Reorientation to LPS, affine registration to T1 template, skull-stripping (BET), resampling to 1x1x1 mm³ isotropic.	Advanced.	3D feature maps processed through attention modules (Channel-wise and Spatial-wise).	Multi-task CNN (Shared backbone with task-specific heads + Residual blocks with attention).	Hybrid (CNN with Multi-task learning + Attention mechanisms).	1. Preprocess 3D MRI. 2. Segmentation using attention-guided U-Net. 3. Extract features via shared 3D CNN backbone. 4. Multi-task prediction (Grading, 1p/19q, IDH) via task-specific FC layers.	10-fold cross-validation.	Not Mentioned.	High Reliability.	Accuracy (LGG/HGG): 95.86%, IDH Status: 91.96%, 1p/19q status: 87.88%.	Limitation: Segmentation and classification are performed as two separate network architectures. Innovation: Multi-task learning (simultaneous 3 tasks); dual-path attention (DPA) mechanism in residual blocks.
17	Preoperative Glioma Grading Based on Hierarchical Information Using Residual Network and Graph Convolutional Network	Fuchun Liu and Mingyuan Li	2023	BraTS 2018 dataset and IXI dataset. Modalities: T1C (T1 contrast-enhanced) and T1.	Multi-modality (Combined).	3D (3D MRI volumes).	Glioma Grading: HGG vs. LGG.	Reorientation (LPS), Affine registration, Bias field correction (N4ITK), Skull-stripping (BET), Normalization [0,1].	Advanced.	Hierarchical feature maps extracted from ResNet layers and transformed into graph topology.	Cascaded ResNet and GCN (Hybrid architecture).	Hybrid.	1. Preprocess 3D MRI data. 2. Encode axial slices using ResNet. 3. Extract hierarchical feature maps at different depths. 4. Construct graph topology from these features. 5. Analyze graph using GCN for final grading classification.	3-fold cross-validation.	Not Mentioned.	High Reliability.	Accuracy: 93.74%, F1-score: 0.907.	Limitation: Use of single modality (T1) in this specific implementation despite MRI being multimodal. Innovation: Using GCN to model the "logical hierarchical relationship" between feature maps, effectively simulating dendritic operations found in brain neuronal structures.

18	Adaptive multi-feature fusion architecture with optimized learning for high-fidelity brain tumor classification in MRI	Mohammed Safy et al	2026	BraTS 2020 dataset. Modalities: T2-weighted and FLAIR.	Multi-modality.	2D (Slices converted from 3D volumes).	Multi-class classification: HG-G, LG-G, and Normal.	Adaptive Gamma Correction (AGC) + Denoising CNN (DnCNN) + Uniform resizing (224x224).	Advanced.	CNN-GLCM Fused Features (CGFFs) - concatenation of flattened deep features and handcrafted GLCM texture descriptors.	Hybrid Ensemble (Transfer Learning CNNs: DenseNet201, EfficientNetB0, NASNetMobile + Stacked Ensemble of RF, XGBoost, LightGBM, SVM).	Hybrid (Fusion of Deep Learning features and handcrafted texture features).	1. Preprocess images (AGC+DnCNN). 2. Extract features using three fine-tuned TRCNNs. 3. Flatten and fuse deep features with GLCM texture descriptors. 4. Classify using SVM and Ensemble models.	Patient-level split.	Not Mentioned.	High Reliability.	Accuracy: 99.05%, Recall: 98.99%, Specificity: 99.52%.	Limitation: Dependence on 2D slice conversion which may lose 3D spatial volumetric context. Innovation: Novel "CGFF" feature fusion strategy capturing both semantic and micro-texture patterns; use of Friedman test for statistical validation.
19	Observing deep radiomics for the classification of glioma grades	Kazuma Kobayashi et al.	2021	BraTS 2019 dataset (T1, Gd-enhanced T1, T2, and FLAIR).	Multi-modality.	2D (Axial slices extracted from 3D volumes).	Glioma Grading: HGG vs. LGG.	Co-registration to anatomical template, resampling to 1mm³, skull-stripping, Z-score normalization.	Advanced.	Vector-quantized latent representations (codebook of feature vectors) summarized as histograms per volume.	Encoder-Decoder CNN with Vector Quantization (VQ-VAE inspired) + Logistic Regression classifier.	Hybrid (Deep feature extraction + Interpretability-focused classification).	1. Train encoder-decoder with Vector Quantization (VQ) on tumor segmentation. 2. Extract latent feature vectors (codebook) as shared imaging markers. 3. Build volume-wise histograms of feature vectors. 4. Classify glioma grades using Logistic Regression based on histograms.	Patient-level split.	Not Mentioned.	High Reliability.	Accuracy: 90%, Specificity: 0.95, Recall: 0.73.	Limitation: No external dataset validation; unclear demographic details of public dataset. Innovation: Introduces "Deep Radiomics"—a data-driven way to fix CNN internal variability using vector quantization to make features shareable and interpretable.
20	Glioma subtype prediction based on radiomics of tumor and peritumoral edema under automatic segmentation	Xiangyu Sun et al.	2024	BraTS 2021 training dataset + Clinical data (Zhongnan Hospital) + UCSF-PDGM (TCIA). Modalities: T1, T1CE, T2, T2-FLAIR.	Multi-modality.	3D (3D MRI volumes, preprocessed to 240x240x155).	Glioma Grading (HGG/LGG) based on WHO 2021 molecular subtypes	Rigid alignment, Co-registration to SRI24 template, Skull-stripping, Isotropic resampling (1mm), Z-score normalization.	Advanced.	Radiomic features (First-order, Shape, Texture matrices) extracted from segmented 3D VOIs of tumor core + edema.	Hybrid: 3D U-Net (for segmentation) + Machine Learning classifiers (RF, SVM, XGBoost, etc.).	Hybrid (Deep Learning-based segmentation + Machine Learning-based classification).	1. Preprocess multimodal 3D MRI. 2. Automated segmentation using 3D U-Net. 3. Extract 96 radiomic features from tumor/edema regions. 4. Classify glioma subtypes using trained ML classifiers (Random Forest).	Patient-level split (Cases were randomly divided into training and validation, and external testing was performed on distinct cases).	Not Mentioned.	High Reliability.	AUC: 0.945 (overall); 0.96 (GBM), 0.914 (Astrocytoma), 0.961 (Oligodendroglioma).	Limitation: Correlation analysis between pathology and morphology was unstable due to outliers; performance varied by tumor category. Innovation: Automated preoperative glioma subtype prediction pipeline based on WHO 2021 classification; demonstrates clinical utility of "whole tumor + edema" radiomics.

21	Interpretable multimodal transformer for prediction of molecular subtypes and grades in adult-type diffuse gliomas	Yunsu Byeon et al.	2025	Institutional set (1053 patients), TCGA (200), UCSF (477). Modalities: T1, T1C, T2, FLAIR.	Multi-modality (Imaging + Clinical data like Age/Sex).	2D (Axial slices selected by tumor area).	Glioma Grading (CNS WHO Grade 2, 3, 4) and Molecular Subtypes (IDH mutation, 1p/19q codeletion).	Isovoxel resampling (1mm³), N4 bias field correction, Co-registration, Skull-stripping (HD-BET), Z-score normalization.	Advanced.	Visual transformer embeddings (for images) + BERT embeddings (for clinical data).	GlioMT (Multimodal Transformer architecture integrating imaging features with LLM-encoded clinical features).	Transformer.	1. Preprocess 3D MRI. 2. Slice selection (based on tumor area). 3. Encode image slices via Visual Transformer. 4. Encode clinical data via BERT. 5. Multimodal fusion via Attention blocks. 6. Multi-task classification head.	Patient-level split.	Not Mentioned.	High Reliability.	AUCs: 0.915-0.981 (IDH mutation), 0.806-0.854 (1p/19q), 0.862-0.960 (Grade).	Limitation: Use of 2D slices instead of 3D volume; lack of advanced sequences (DWI/DSC). Innovation: GlioMT framework; integration of LLM (BERT) for clinical data; provides interpretability via attention maps and modality contribution scores.
22	A novel image signature-based radiomics method to achieve precise diagnosis and prognostic stratification of gliomas	Huigao Luo et al.	2020	655 clinical MRI scans from Huashan Hospital & Shanghai International Medical Center. Modalities: T1C (T1 contrast) and T2-FLAIR.	Multi-modality	3D (32x32x32 patches for training, full 3D ROI for feature extraction)	Histopathological diagnosis (GBM, Astrocytoma, Oligodendroglioma) and molecular subtyping (IDH, 1p19q status) directly mapping to Glioma Grading (HGG vs LGG).	Skull-stripping and intensity normalization	Advanced	Deep network features ("Image Signature") extracted from the bottom layer of a 3D U-Net, followed by Fisher vector encoding for dimension reduction.	3D U-Net (for segmentation and deep feature extraction) + Sparse Representation Classification.	Hybrid (Deep learning for feature extraction + Machine learning/Sparse representation for classification).	1. Preprocess 3D MRI (skull stripping, normalization). 2. Train 3D U-Net for tumor segmentation. 3. Extract deep network features from the bottom layer of the trained 3D U-Net. 4. Reduce feature dimension using Fisher vector clustering and t-test. 5. Classify histological and molecular subtypes using sparse representation classifiers.	Patient-level split (Strict separation into independent training, cross-validation, and independent testing cohorts).	Not Mentioned	High Reliability	Accuracy: 99.8% (Binary), Macro-F1: 0.998.	Limitation: 2D-based approach loses volumetric context; high computational cost of the dual-stream attention. Innovation: Introduces "Deformable Cross-Modal Attention" (DCMA); frequency-aware learning for texture discrimination; robust cross-dataset generalization.
23	Glioma Grading Prediction by Exploring Radiomics and Deep Learning Features	Taohui Xiao et al.	2019	BraTS 2018 dataset (285 subjects). Modalities: T1ce (T1-weighted contrast-enhanced)	Multi-modality (Combines Radiomics features + Deep Learning features)	3D (Extracted from 3D volumes).	Glioma Grading (HGG vs LGG)	Skull-stripping, Interpolation (1mm isotropic voxel size), Normalization	Advanced	Combined feature subsets: Radiomics (shape, texture) + Deep features (from VGG-16)	Hybrid (VGG-16 for deep features + RFE for selection + SVM/LR/LDA classifiers)	Hybrid	1. Tumor segmentation (manual/manual-corrected). 2. Extraction of radiomics features (shape/texture) and Deep features (via fine-tuned VGG-16). 3. Feature subset selection using Recursive Feature Elimination (RFE). 4. Training of classifiers (LR, SVM, LDA).	Not explicitly mentioned, but implies standard cohort splitting	Not Mentioned	Low Reliability	AUC 94.4% (Combined method)	Innovation: Integration of Radiomics + Deep Learning features for better representation. Limitation: Reliance on VGG-16 fine-tuning (older architecture); relatively small cohort from BraTS 2018.

24	Glioma classification in MRI using a hybrid deep learning framework with majority vote ensemble	Sonam Saluja, Munesh Chandra Trivedi	2025	BraTS 2018 and external validation on BraTS 2020 Imaging Type: T2-Weighted MRI (T2-W MRI)	Single-modality	2D MRI slices	Classification of glioma into HGG vs LGG	Image resizing, normalization, augmentation, fine-tuning preparation for CNN models	Basic	Single-channel T2 MRI images represented as CNN input tensors	Fine-tuned pre-trained CNNs (AlexNet, VGG16, SqueezeNet, GoogLeNet, ResNet50) combined using Majority Voting, Weighted Voting, and Stacked Ensemble	Ensemble	1. Acquire T2 MRI images from BraTS datasets 2. Apply preprocessing and augmentation 3. Fine-tune five pre-trained CNN models 4. Generate predictions from individual CNNs 5. Combine predictions using MJ, WV, and SE ensemble methods 6. Evaluate performance using Accuracy, Sensitivity, Specificity, and AUC	External validation on BraTS 2020 suggests an attempt to improve generalization; exact patient-level split strategy not clearly specified	Not Mentioned	Low Reliability	BraTS 2018: Accuracy 99.35%, Sensitivity 99.50%, Specificity 99.45%, AUC 99.40% using Stacked Ensemble BraTS 2020: Accuracy 97.90%, Sensitivity 98.05%, Specificity 97.80%, AUC 97.85%	Limitation: Uses only single-modality T2 MRI and does not clearly describe patient-level data splitting or calibration methods. Innovation: Proposes a hybrid ensemble framework combining five fine-tuned CNNs using Majority Voting, Weighted Voting, and Stacked Ensemble with strong external validation on BraTS 2020.
25	SEMD-Net: A transformer based approach for brain tumor segmentation and classification	S. Vidhya, P.M. Siva Raja, R.P. Sumithra, Moses Garuba, Xiao-Zhi Gao	2026	BraTS 2020 and BraTS 2021 Imaging Type: Multi-modal MRI images	Multi-modality	Likely 3D MRI volumes	Brain tumor segmentation and glioma grade classification (HGG vs LGG)	Multi-modal MRI preparation, progressive transfer learning, five-fold cross-validation setup	Advanced	Multi-branch feature representation combining spatial-channel attention and transformer-based embeddings	SEMD-Net combining Squeeze-and-Excitation, MAMBA, DeiSwin++ Transformer, and progressive transfer learning	Hybrid	1. Input multi-modal MRI scans from BraTS 2020/2021 2. Preprocess and organize MRI modalities 3. Extract global and local tumor features using multi-branch SEMD-Net 4. Apply spatial-channel attention and transformer representation learning 5. Perform tumor segmentation 6. Perform HGG vs LGG classification 7. Evaluate using five-fold cross-validation	Five-fold cross-validation was used, but explicit patient-level data splitting was Not Mentioned	Not Mentioned	Low Reliability	Segmentation: DSC = 0.88 ± 0.02, IoU = 0.86 ± 0.03, HD = 3.5 ± 0.8 mm. Classification: Accuracy = 97.5%, Precision = 98.9%, Recall = 98.8%, F1-score = 98.85%	Limitation: External dataset validation is still ongoing and patient-level leakage prevention was not clearly described. Innovation: Introduces SEMD-Net, integrating MAMBA, DeiSwin++ Transformer, spatial-channel attention, and progressive transfer learning for unified segmentation and HGG/LGG classification.